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Rooth-Partee conditionals and the symmetry of disjunction

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Abstract: It is argued that *or* is inherently symmetric, based on the intuitive equivalence of *If x is P or Q , then y is* and *If x is Q or P , then y is* – referred to as Rooth-Partee conditionals. As observed in Rooth and Partee [1982. Conjunction, type ambiguity and wide scope *or*. In Daniel P. Flickinger, Marlys Macken & Nancy Wiegand (eds.), *Proceedings of WCCFL*, vol. 1], the elided phrase in the consequent clause of these conditionals brings about a reading paraphrased as *If x is P , y is P , and if x is Q , y is Q* . Accounting for that reading proves to be problematic for asymmetric approaches to *or*, even those that rely on presupposition accommodation.

Keywords: conditionals; symmetry; disjunctive antecedents; presupposition accommodation

1 The problem: Is *or* inherently symmetric?

This paper responds to theories of presupposition projection which – implicitly or explicitly – endorse the universal principle *CON*-asymmetry in (1), in line with Karttunen (1974). *CON*-asymmetry says that all natural language connectives require – in virtue of their meaning – that the presupposition of their leftmost clause be a presupposition of the embedding clause.

(1) *CON*-asymmetry

For any p_1 and p_2 , connective *CON*, and $q \in \{\text{CON } p_1, p_1 \text{ CON } p_2, \text{CON } p_1 p_2\}$ such that q is well-formed: q presupposes what p_1 presupposes (and if $q \in \{p_1 \text{ CON } p_2, \text{CON } p_1 p_2\}$, q does not necessarily presuppose what p_2 presupposes).

That the meaning of the negating connective *I don't think that* is compatible with *CON*-asymmetry is shown by the fact that (3) is a potentially felicitous response to both (2a) and (2b). (3) contests *Mia has money*, which is presupposed by both (2a) and (2b). (We ignore any reading where *her* is not “bound” by *Mia*.)

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- (2) a. Mia is proud of her money.
- b. I don't think that Mia is proud of her money.
- (3) I can neither agree nor disagree with you, because I don't think Mia has any money at all.

By contrast, (4) is a contradictory response to (2a/b). This confirms that what is contested in (3) is not what is asserted in (2a/b).

- (4) ⊥I can neither agree nor disagree with you, because I don't think that Mia is proud of her money – she is ashamed of it.

That the meanings of *and* and *if* are compatible with *CON*-asymmetry is shown by the potential felicity of (5a,c,d), which contrasts with the inherent oddity of (5b): according to *CON*-asymmetry, (5b) and (5d) presuppose *Mia has money*. However, *and she has money* is redundant in (5b). Neither (5a) nor (5c) presupposes *Mia has money*.

- (5) a. If Mia has money and she is proud of her money, Sue is jealous.
- b. #If Mia is proud of her money and she has money, Sue is jealous.¹
- c. If Mia has tons of money and she is proud of her money, Sue is jealous.
- d. If Mia is proud of her money and she has tons of money, Sue is jealous.

That *and* and *if* are compatible with *CON*-asymmetry is also supported by the fact that (3) is a potentially felicitous response to (5d), but an inherently odd response to (5a) and (5c). In addition, (6) is a contradictory response to (5d).

- (6) ⊥I can neither agree nor disagree with you, because I think that if Mia is proud of her money and she has a lot of money, Sue is not jealous – she's disgusted.

At a first glance, the behavior of *or* seems to be incompatible with *CON*-asymmetry (see, for example, Karttunen and Peters (1979) and Soames [1979]), as illustrated by the intuitive equivalence of (7a) and (7b), and by the fact that (3) is an odd response to both of them.

- (7) a. (Either) Mia has no money or she is proud of her money.
- b. (Either) Mia is proud of her money or she has no money.

However, for the sake of generality and explanatory adequacy, it seems justified to say that *CON*-asymmetry holds, and to attribute the fact that (7b) does not mimic (5b)

¹ One might argue that the oddity of the unembedded #*Mia is proud of her money and she has money* is due, not to what *and* presupposes, but rather to the processing of p_1 and p_2 as two consecutive main clauses (see Chemla & Schlenker [2012]). Importantly, this is not an available option for the antecedent clause of (5b).

to an independent mechanism that overrides **CON**-asymmetry. Many scholars who take that position assume that the mechanism in question is presupposition accommodation. The intuitive equivalence of (7a) and (7b), on that view, is due to the accommodation of *Mia has money* “below” *or* in (7b).

Despite the explanatory appeal of **CON**-asymmetry, we argue that accommodation does not account for the intuitive symmetry of *or*. Our argument is based on conditional sentences where the antecedent clause is disjunctive and the verb phrase in the consequent clause is elided. Such conditional sentences are first discussed in Rooth and Partee (1982), and we refer to them as *Rooth-Partee conditionals*. The intuitively equivalent (8a) and (8b) are examples.

- (8) a. If Mia is (either) penniless or proud of her money, then Sue is (too).
- b. If Mia is (either) proud of her money or penniless, then Sue is (too).

One of the “bound” readings of (8a,b) asserts that if Mia is penniless, Sue is penniless, and if Mia has money and is proud of her money, Sue has money and is proud of her own money. That reading often presupposes *If Mia has money, Sue has money* (as observed in Abenina-Adar and Sharvit 2021). We argue that an asymmetric *or* fails to predict this, with or without presupposition accommodation.

A word of caution before we go any further. While all the consultants in our informal survey report the judgments described above, there is no consensus in the literature regarding what the connectives presuppose, intuitively or formally (see, for example, Rothschild (2011/2015), Chemla and Schlenker (2012), Mandelkern et al. (2020), Kalomoiros and Schwarz (2024)). Given this, the readers are asked to read this paper as a response to the following view: (i) *and* and *if* are intuitively asymmetric, (ii) *or* is intuitively symmetric, and (iii) **CON**-asymmetry holds, and (iv) the intuitive symmetry of *or* is a bi-product of presupposition accommodation.²

The rest of the paper is organized as follows. Section 2 shows how combining **CON**-asymmetry with optional accommodation accounts for the intuitive symmetry of *or* in unembedded sentences. Sections 3–4 show that this combination fails to account for Rooth-Partee conditionals. Section 5 presents an alternative theory of the connectives that accounts for Rooth-Partee conditionals. Section 6 discusses some additional consequences.

2 There seem to be no speakers (not even in our survey) for whom (1a) and (1b) below are equivalent. Contrasts such as this seem to inform us more about how definite descriptions and other presupposition triggers are processed, than about the lexical meaning of *or*. For discussion, see Roberts and Simons (2024).

- (I) a. Either Jones is in a meeting with a new student, or else the student is waiting in the lab.
- b. Either the student is waiting in the lab, or else Jones is in a meeting with a new student.

2 Redundancy contrasts and accommodation

Given (5), it is unsurprising that (9b) and (10b) are inherently infelicitous, while (9a) and (10a) are potentially felicitous. The potentially felicitous (7a) and (7b) are repeated in (11).

- (9) a. If Mia has money, she is proud of her money.
b. #If Mia is proud of her money, she has money.
- (10) a. Mia has money and she is proud of her money.
b. #Mia is proud of her money and she has money.
- (11) a. (Either) Mia has no money or she is proud of her money.
b. (Either) Mia is proud of her money or she has no money.

We refer to the contrasts illustrated by (9)–(11) as “redundancy contrasts”.

From a purely descriptive perspective, *if*, *and* and negation seem to inherit the presuppositions of their leftmost argument but *or* does not. Thus, at a purely descriptive level, the redundancy contrasts in (9)–(11) are captured by (12)–(13) in combination with a neo-Gricean version of Grice’s Cooperative Principle (see Grice (1989)) – henceforth CP – which subsumes (14).

- (12) a. For all q in $\{if\ p_1\ p_2, p_1\ and\ p_2, neg\ p_1\}$, q presupposes what p_1 presupposes.
b. $p_1\ or\ p_2$ presupposes at least one p in $\{\text{what } p_1 \text{ presupposes, what } p_2 \text{ presupposes}\}$.
- (13) a. *if* $p_1\ p_2$ asserts that at least one p in $\{neg\ p_1, p_2\}$ is true.
b. $p_1\ and\ p_2$ asserts that every p in $\{p_1, p_2\}$ is true.
c. $p_1\ or\ p_2$ asserts that at least one p in $\{p_1, p_2\}$ is true.
d. *neg* p_1 asserts that p_1 is false.
- (14) p is felicitous only if p has a truth value and contains no uninformative parts.

Given that (9b) and (10b) presuppose *Mia has money*, the rightmost clause in (9b) and (10b) – namely, *she has money* – is always uninformative in (9b) and (10b). Accordingly, (9b) and (10b) are both inherent violators of CP. By contrast, (9a) and (10a) are potentially CP-compliant. On the other hand, neither (11a) nor (11b) is an inherent violator of CP; moreover, they are equivalent to each other. If *or* had the same presupposition as *if* and *and*, (11b) would not be equivalent to (11a) (and would – like (9b) and (10b) – violate CP, because it would presuppose *Mia has money*, making its rightmost clause uninformative).

What theory of the connectives predicts (12)–(13)? One possibility is that (12)–(13) are the actual formal meanings of the connectives. This possibility is obviously

unsatisfactory, as it does not explain why *or* – unlike *neg*, *if* and *and* – does not project the presuppositions of its leftmost clause. A conceptually more appealing possibility is that **CON**-asymmetry holds, as in Karttunen (1974). An obvious challenge for that theory is the potential felicity of (11b). A popular way of addressing this challenge involves supplementing **CON**-asymmetry with presupposition accommodation. According to the received wisdom, upon hearing some p that presupposes q , cooperative interlocutors often globally accommodate q – i.e., they treat q as true – even if, prior to hearing p , they were agnostic about q (see, for example, Karttunen (1974), Stalnaker (1974), Lewis (1979)). Adopting insights from Gazdar (1979), Heim (1982, 1983), Geurts (1996, 1999), Beaver and Krahmer (2001), Schlenker (2008, 2009, 2011), Hirsch and Hackl (2014) and others, many scholars contend that presuppositions can be accommodated both globally and non-globally (i.e., in embedded positions). Let us discuss a specific toy theory, which we call $K+$, that accounts for redundancy contrasts by combining **CON**-asymmetry with such liberal accommodation.

We assume that when **CON** is bi-clausal, the L(ogical)F(orm) is $[[\text{CON} \wedge \phi] \wedge \phi]$, even when surface word order is different. LFs are interpreted by the interpretation function $\llbracket \cdot \rrbracket$, relativized to a world and a context (where context $c \supset \{w_c\}$, w_c is the world of c , and if c is an utterance context, c also includes the interlocutors of the utterance). $\wedge \phi$ and $\wedge \phi$ are interpreted by (15).³ As (16) implies, the sentential connectives of $K+$ presuppose only what is required by **CON**-asymmetry, which is restated in (17).⁴

$$(15) \quad \text{For any LF } \delta, \llbracket \wedge \delta \rrbracket^{w,c} = [\lambda w': \delta \in \text{Dom}(\llbracket \cdot \rrbracket^{w',c}). \llbracket \delta \rrbracket^{w',c}]$$

(16) If defined:

- a. $\llbracket \text{if}^{K+} \rrbracket^{w,c}(p_1) = [\lambda p_2. p_1(w) = \text{False} \text{ or } p_2(w) = \text{True}]$
- b. $\llbracket \text{and}^{K+} \rrbracket^{w,c}(p_1) = [\lambda p_2. p_1(w) = p_2(w) = \text{True}]$
- c. $\llbracket \text{or}^{K+} \rrbracket^{w,c}(p_1) = [\lambda p_2. p_1(w) = \text{True} \text{ or } p_2(w) = \text{True}]$
- d. $\llbracket \text{neg}^{K+} \rrbracket^{w,c}(p_1) = \text{True} \text{ iff } p_1(w) = \text{False}$

(17) **CON**-asymmetry $\approx$ (1)

$\llbracket \text{CON} \rrbracket^{w,c}(p_1)$ is defined iff $p_1(w)$ is defined (i.e., iff $p_1(w) \in \{\text{True}, \text{False}\}$).

3 '[λU : U is of type σ , and A, B]' is read as either (I) or (II) below, whichever is applicable (see Heim and Kratzer (1998)). ' U is of type σ ' is omitted when σ is determined by convention; A is omitted when trivially true.

(I) the smallest function that maps every U of type σ such that A to B .

(II) the smallest function that maps every U of type σ such that A to True, if B , and to False otherwise.

4 The assumption that the arguments of **CON** are propositions (i.e., functions from worlds to truth values), rather than truth values, is justified partly in Section 3 and more generally in Section 5. For now, it is merely harmless.

For transparency, we represent accommodation syntactically, as conjunction. Accordingly, (18a) has (at least) the LFs in (19), and (18b) has (at least) the LFs in (20). (19a) and (20a) are accommodation-free, in (19b) and (20b) $\wedge$ [*Mia has money*] is accommodated non-globally, and in (19c) and (20c) $\wedge$ [*Mia has money*] is accommodated globally.

- (18) a. #If Mia is proud of her money, she has money.
 b. #Mia is proud of her money and she has money.
- (19) a. $\text{if}^{K+} \wedge$ [*Mia proud of Mia's money*] $\wedge$ [*Mia has money*]
 b. $\text{if}^{K+} \wedge$ [$\text{and}^{K+} \wedge$ [*Mia has money*] $\wedge$ [*Mia proud of Mia's money*]] $\wedge$ [*Mia has money*]
 c. $\text{and}^{K+} \wedge$ [*Mia has money*] $\wedge$ [$\text{if}^{K+} \wedge$ [*Mia proud of Mia's money*] $\wedge$ [*Mia has money*]]
- (20) a. $\text{and}^{K+} \wedge$ [*Mia proud of Mia's money*] $\wedge$ [*Mia has money*]
 b. $\text{and}^{K+} \wedge$ [$\text{and}^{K+} \wedge$ [*Mia has money*] $\wedge$ [*Mia proud of Mia's money*]] $\wedge$ [*Mia has money*]
 c. $\text{and}^{K+} \wedge$ [*Mia has money*] $\wedge$ [$\text{and}^{K+} \wedge$ [*Mia proud of Mia's money*] $\wedge$ [*Mia has money*]]

CP now applies to LFs, not surface sentences, and subsumes (21)–(22). EP_w is the set of worlds compatible with the relevant body of knowledge in w . If $w = w_c$ and c is an utterance context, EP_w is the set of worlds compatible what the interlocutors of c agree on in w . Since δ dominates itself, it follows that when δ is declarative, EP_{w_c} contains both $\wedge\delta$ -worlds and $\wedge[\text{neg}^{K+} \wedge \delta]$ -worlds.

- (21) LF δ is CP-compliant relative to utterance context c only if:
- $\text{EP}_{w_c} \subseteq \{w \mid \llbracket \delta \rrbracket^{w,c} \text{ is defined}\}$,
 - there is no ε such that δ dominates ε and ε is uninformative in δ relative to c .
- (22) ε is uninformative in δ relative to c iff: ε is dominated by δ , and there is a $\pi \in \{\text{tr}, \text{fl}\}$ such that for all $w \in \text{EP}_{w_c}$, $\llbracket \delta \rrbracket^{w,c} = \llbracket \delta_{[\varepsilon/\pi]} \rrbracket^{w,c}$, where:
- either (I) ε is not an occurrence of π , in which case $\delta_{[\varepsilon/\pi]}$ is just like δ except that ε is replaced with an occurrence of π ,
 or (II) ε is an occurrence of π , in which case $\delta_{[\varepsilon/\pi]} = \delta$, and
 - for all (w, c') : $\llbracket \text{tr} \rrbracket^{w,c'} = \text{True}$ and $\llbracket \text{fl} \rrbracket^{w,c'} = \text{False}$.

When $\text{EP}_{w_c} \subseteq \{w \mid \llbracket (19a) \rrbracket^{w,c} \in \{\text{True}, \text{False}\}\}$ (i.e., $\text{EP}_{w_c} \subseteq \{w \mid \text{Mia has money in } w\}$), at least one sub-LF of (19a) is uninformative in (19a) (for example, [*Mia has money*] can be replaced with *tr* without affecting the meaning). As a result, (19a) is an inherent

violator of CP. For the same reason, (19b,c) and (20a,b,c) are all inherent violators of CP. In fact, all the syntactically well-formed LF of (18a) and (18b) are inherent violators of CP; this explains the inherent oddity of (18a) and (18b). By contrast, (23a) and (23b) are potentially felicitous, as (24a) and (24b) are potentially CP-compliant: there are contexts *c* where no sub-LF of (24a) or (24b) can be replaced with either *tr* or *fl* without affecting the meaning.

- (23) a. If Mia has money, she is proud of her money.
 b. Mia has money and she is proud of her money.
- (24) a. $if^{K+} \wedge [Mia \text{ has money}] \wedge [Mia \text{ proud of Mia's money}]$
 b. $and^{K+} \wedge [Mia \text{ has money}] \wedge [Mia \text{ proud of Mia's money}]$

Likewise, (25a) and (26a) are potentially felicitous as they have, respectively, the LF in (25b) and (26b), which are potentially CP-compliant. Moreover, thanks to the non-global accommodation of $\wedge [Mia \text{ has money}]$ in (26b), (25b) and (26b) are equivalent to each other.

- (25) a. (Either) Mia has no money or she is proud of her money.
 b. $or^{K+} \wedge [Mia \text{ has no money}] \wedge [Mia \text{ proud of Mia's money}]$
- (26) a. (Either) Mia is proud of her money or she has no money.
 b. $or^{K+} \wedge [and^{K+} \wedge [Mia \text{ has money}] \wedge [Mia \text{ proud of Mia's money}]] \wedge [Mia \text{ has no money}]$

Notice that (27) – an inherent violator of CP – is also an LF of (26a).

- (27) $or^{K+} \wedge [Mia \text{ proud of Mia's money}] \wedge [Mia \text{ has no money}]$

Since (27) presupposes *Mia has money*, it is not equivalent to (25b) or to (26b) (neither of which presupposes *Mia has money*). Moreover, $[Mia \text{ has no money}]$ is uninformative in (27), as replacing it with *fl* has no effect on the meaning. Crucially, (26b) is always an available LF of (26a), and so, (26a) – unlike (18a) and (18b) – is potentially felicitous.

Since $K+$ derives redundancy contrasts from *CON*-asymmetry and general principles, it has a clear conceptual appeal (though *CON*-asymmetry itself is stipulated; see Soames [1989])⁵. Nevertheless, as we show below, $K+$ is not tenable, because it cannot – in principle – account for the intuitive symmetry of *or* in antecedents of Rooth-Partee conditionals (i.e., conditionals that have a disjunctive antecedent and a partly elided VP in the consequent).

⁵ Our goal in this section is merely to show that accommodation+*CON*-asymmetry+CP can, in principle, account for redundancy contrasts. For more fine-grained renditions of CP (which account for a much wider range of facts) see, for example, Schlenker (2008, 2009, 2011).

3 Rooth-Partee conditionals: A challenge for K⁺

A problem arises for K⁺ in some cases where a disjunction is embedded in the antecedent of a conditional. K⁺ correctly predicts (28a,b) – which are plain conditionals – to be intuitively equivalent, but fails to predict that the Rooth-Partee conditionals (29a,b) are equivalent to each other, because it fails to generate the presuppositions associated with the ellipsis site.

- (28) a. If Mia is (either) penniless or proud of her money, then Sue is unhappy.
 b. If Mia is (either) proud of her money or penniless, then Sue is unhappy.
- (29) a. If Mia is (either) penniless or proud of her money, then Sue is (too).
 b. If Mia is (either) proud of her money or penniless, then Sue is (too).

We discuss simpler cases of Rooth-Partee conditionals in 3.1. We come back to (29) in 3.2.

3.1 Basic Rooth-Partee conditionals

Consider the Rooth-Partee conditional in (30). We assume, merely for simplicity (and until Section 5) that both *bored* and *penniless* are presupposition-free.⁶

- (30) If Mia is bored or penniless, then Sue is (too).

As observed in Rooth and Partee (1982), (30) is ambiguous: it has the *if-over- $\exists$* reading paraphrased in (31) and the *$\forall$ -over-if* reading, paraphrased in (32). Both readings are paraphrased as a presupposition-free conjunction of two plain conditionals. In (31), the consequent-VP is *bored or penniless* in the two conjoined plain conditionals. In (32), the consequent-VP matches the antecedent-VP in the two conjoined plain conditionals.

- (31) Assertion of the *if-over- $\exists$* reading of (30)
 If Mia is bored, Sue is bored or penniless, and
 if Mia is penniless, Sue is bored or penniless.
- (32) Assertion of the *$\forall$ -over-if* reading of (30)
 If Mia is bored, Sue is bored, and
 if Mia is penniless, Sue is penniless.

That the two readings are indeed distinct readings (rather than *$\forall$ -over-if* being a special case of *if-over- $\exists$*) is shown by: (i) the fact that both (33a) and (33b) can be

⁶ This is clearly incorrect. For example, *penniless* presupposes *can, in principle, have money*.

felicitous responses to (30), and (ii) the fact that the plain conditional (34) – which has no ellipsis in the consequent – has only the *if*-over- $\exists$ reading. Indeed, (33a) can be a felicitous response to (34), but (33b) is a contradictory response to (34).

- (33) a. I don't think so. Mia is bored, but Sue has money and she is not bored.
(implies that both readings of (30) are false)
b. I don't think so. Mia is bored, but Sue is penniless and not bored.
(implies that the $\forall$ -over-*if* reading of (30) is false)
- (34) If Mia is bored or penniless, then Sue is bored or penniless.

How does the ambiguity of (30) come about? To answer this question, it is instructive to consider (35), which – like (30) – has a disjunctive antecedent. Unlike (30), (35) has an overt pronoun in the consequent.

- (35) If Mia caught a fish or Sue trapped a rabbit, Fred cooked it.

As observed in Stone (1992), (35) is most naturally interpreted as a conjunction of the two plain conditionals in (36). The pronoun *it*, which carries singular morphology, is an E-type pronoun that denotes either the individual made salient in virtue of being “mentioned” in the antecedent of the first inferred plain conditional, or the individual made salient in virtue of being “mentioned” in the antecedent of the second inferred plain conditional.

- (36) If Mia caught a fish, Fred cooked the fish that Mia caught ($\approx it$),
and
if Sue trapped a rabbit, Fred cooked the rabbit that Sue trapped ($\approx it$)

In Stone (1992), Rooth-Partee conditionals such as (30) are thought of as instantiating E-type anaphora along the lines of Heim (1990). The core idea, on this view, is that the ellipsis site in the consequent is a phonetically unrealized E-type pronoun that denotes a property made salient in virtue of being “mentioned” in the antecedent; that pronoun may be interpreted in more than one way. We execute this idea with a simplified version of Stone (1992), which we call K^{++} .

K^{++} is just like K^{+} except that **or** $^{K^{++}}$ and **and** $^{K^{++}}$ are type-flexible (as in Partee and Rooth (1982) and Rooth and Partee (1982)), **if** $^{K^{++}}$ is type-rigid and intensional, and **CON**-asymmetry holds. More concretely, **if** $^{K^{++}}$ has the type-rigid intensional meaning in (37), where the assertion of **if** $^{K^{++}}$ requires all p_1 -worlds in EP_w to be p_2 -worlds. Assuming (38), **and** $^{K^{++}}$ and **or** $^{K^{++}}$ have the type-flexible non-intensional meanings in (39)–(40).

- (37) $\llbracket \mathbf{if}^{K^{++}} \rrbracket^{w,c}(p_1)$ is defined iff $p_1(w) \in \{\text{True}, \text{False}\}$.
If defined, $\llbracket \mathbf{if}^{K^{++}} \rrbracket^{w,c}(p_1) = [\lambda p_2. \{w' \mid w' \in EP_w \text{ and } p_1(w') = \text{True}\} \subseteq \{w' \mid p_2(w') = \text{True}\}]$.

- (38) For any (Z, x) such that Z is a property of individuals and x is an individual, $Z^x = [\lambda w': Z(w')(x) \in \{\text{True}, \text{False}\}. Z(w')(x)]$.
- (39) When P_1 is a proposition, $\llbracket \text{and}^{K++} \rrbracket^{w,c}(P_1)$ is defined iff $\llbracket \text{if}^{K++} \rrbracket^{w,c}(P_1)$ is defined.
 If defined, $\llbracket \text{and}^{K++} \rrbracket^{w,c}(P_1) = [\lambda P_2. P_1(w) = P_2(w) = \text{True}]$.
 When P_1 is a property of individuals, $\llbracket \text{and}^{K++} \rrbracket^{w,c}(P_1) = [\lambda P_2. [\lambda x: \llbracket \text{and}^{K++} \rrbracket^{w,c}(P_1^x)(P_2^x)$ is defined. $\llbracket \text{and}^{K++} \rrbracket^{w,c}(P_1^x)(P_2^x)]$.
- (40) When P_1 is a proposition, $\llbracket \text{or}^{K++} \rrbracket^{w,c}(P_1)$ is defined iff $\llbracket \text{if}^{K++} \rrbracket^{w,c}(P_1)$ is defined.
 If defined, $\llbracket \text{or}^{K++} \rrbracket^{w,c}(P_1) = [\lambda P_2. P_1(w) = \text{True} \text{ or } P_2(w) = \text{True}]$.
 When P_1 is a property of individuals, $\llbracket \text{or}^{K++} \rrbracket^{w,c}(P_1) = [\lambda P_2. [\lambda x: \llbracket \text{or}^{K++} \rrbracket^{w,c}(P_1^x)(P_2^x)$ is defined. $\llbracket \text{or}^{K++} \rrbracket^{w,c}(P_1^x)(P_2^x)]$.

Accordingly, (30) has the LF in (41a), which has the general schematic assertion in (41b). The meanings of *bored* and *penniless* are given in (42). $\forall \text{prop}_E$ is interpreted by (43).

- (41) a. $\text{if}^{K++} \wedge [\text{Mia} [\text{or}^{K++} \wedge \text{bored} \wedge \text{penniless}]] \wedge [\text{Sue} \forall \text{prop}_E]$
 b. $\{w' \in \text{EP}_w \mid \text{Mia is bored or penniless in } w'\} \subseteq \{w' \mid \llbracket \text{prop}_E \rrbracket^{w',c}(w')(\text{Sue}) = \text{True}\}$.
- (42) a. $\llbracket \text{bored} \rrbracket^{w,c} = [\lambda x. x \text{ is bored in } w]$
 b. $\llbracket \text{penniless} \rrbracket^{w,c} = [\lambda x. x \text{ has no money in } w]$
- (43) $\forall \delta \in \text{Dom}(\llbracket \cdot \rrbracket^{w,c})$ iff $\llbracket \delta \rrbracket^{w,c}$ is a function such that $w \in \text{Dom}(\llbracket \delta \rrbracket^{w,c})$, in which case:
 $\llbracket \forall \delta \rrbracket^{w,c} = \llbracket \delta \rrbracket^{w,c}(w)$.

It remains to determine how prop_E is interpreted. Generalizing Heim's (1990) theory of E-type anaphora, prop_E refers to the property made salient by the antecedent of (41a). Stone (1992) proposes that there are two possible interpretations for prop_E . Slightly modifying Stone's proposal, we provide these two options in (44) – f1prop_E and f2prop_E . Accordingly, the schematic meaning in (41b) yields the two possible meanings in (45): the meaning in (45a) accounts for the *if-over- $\exists$* reading of (30), and the meaning in (45b) accounts for its *$\forall$ -over-if* reading.⁷

7 In Stone (1992), as in Heim (1990), *if* quantifies over minimal situations, not whole worlds. Our simplifying assumption that if^{K++} quantifies over whole worlds is good enough for current purposes.

- (44) a. **prop_E** denotes $f1prop_E = \llbracket \wedge [or^{K++} \wedge bored \wedge penniless] \rrbracket^{w',c}$
 b. **prop_E** denotes $f2prop_E$, which is such that for all w :
 if $\llbracket bored \rrbracket^{w,c}(Mia) = \text{True}$ and $\llbracket penniless \rrbracket^{w,c}(Mia) = \text{False}$,
 $f2prop_E(w) = [\lambda x. \llbracket bored \rrbracket^{w,c}(x) = \text{True}$ and $\llbracket penniless \rrbracket^{w,c}(x) = \text{False}]$,
 if $\llbracket bored \rrbracket^{w,c}(Mia) = \text{False}$ and $\llbracket penniless \rrbracket^{w,c}(Mia) = \text{True}$,
 $f2prop_E(w) = [\lambda x. \llbracket bored \rrbracket^{w,c}(x) = \text{False}$ and $\llbracket penniless \rrbracket^{w,c}(x) = \text{True}]$,
 if $\llbracket bored \rrbracket^{w,c}(Mia) = \llbracket penniless \rrbracket^{w,c}(Mia) = \text{True}$,
 $f2prop_E(w) = [\lambda x. \llbracket bored \rrbracket^{w,c}(x) = \llbracket penniless \rrbracket^{w,c}(x) = \text{True}]$.
- (45) a. if-over- $\exists$ meaning of (30)–(41a) paired with $f1prop_E$:
 For all $w' \in EP_w$ such that Mia is bored or penniless in w' ,
 Sue is bored or penniless in w' .
 b. $\forall$ -over-if meaning of (30)–(41a) paired with $f2prop_E$:
 For all $w' \in EP_w$ such that Mia is bored in w' , Sue is bored in w' , and
 for all $w' \in EP_w$ such that Mia is penniless in w' , Sue is penniless in w' .

The ellipsis-free counterpart of (30) in (34) lacks the $\forall$ -over-if reading, because that reading crucially depends on the ambiguity of **prop_E**, yet (34) has only the **prop_E**-less LF in (46).

- (46) $if^{K++} \wedge [Mia [or^{K++} \wedge bored \wedge penniless]] \wedge [Sue [or^{K++} \wedge bored \wedge penniless]]$

In addition, both (47a) and (47b) – which have a conjunctive antecedent – lack a $\forall$ -over-if reading. The LF of (47b) has the pre-requisite **prop_E**, but the antecedent – by assumption – supplies only one interpretive option (namely, $\llbracket \wedge [and^{K++} \wedge bored \wedge penniless] \rrbracket^{w',c}$).

- (47) a. If Mia is bored and penniless, then Sue is bored and penniless.
 b. If Mia is bored and penniless, then Sue is (too).

Equipped with this analysis of the ambiguity of Rooth-Partee conditionals, let us turn to cases where at least one of the disjuncts is presuppositional.

3.2 Rooth-Partee conditionals with at least one presuppositional disjunct

Consider (48), where *proud of her money* presupposes *has money*. (48) has an if-over- $\exists$ reading and a $\forall$ -over-if reading. (We ignore any reading where *she/her* is referential rather than bound.)

- (48) If Mia is (either) penniless or proud of her money, then Sue is (too).

The ambiguity of (48) is illustrated by the fact that both (49a) and (49b) can be felicitous responses to (48), but only (49a) can be a felicitous response to (50), with no ellipsis in the consequent. (50) has only the *if-over- $\exists$* reading.

- (49) a. I don't think so; Mia is penniless, Sue has money, and Sue is ashamed of her money.
 b. I don't think so; both Mia and Sue have money, Mia is proud of her money, and Sue is ashamed of hers.
- (50) If Mia is penniless or proud of her money, then Sue is penniless or proud of her money.

The *if-over- $\exists$* reading of (48), paraphrased in (51), is presupposition-free. The $\forall$ -over-*if* reading is presuppositional. The presupposition of the $\forall$ -over-*if* reading is quite complex; we temporarily simplify it, as in (52) (see Section 4 for a more detailed and accurate description of that presupposition).

- (51) Assertion of *if-over- $\exists$* reading of (48)
 If Mia is penniless, Sue is penniless or proud of Sue's money, and
 if Mia has money and is proud of Mia's money, Sue is penniless or proud of Sue's money.
- (52) a. Presupposition of the $\forall$ -over-*if* reading of (48)
 If Mia has money, Sue has money.
 b. Assertion of $\forall$ -over-*if* reading of (48)
 If Mia is penniless, Sue is penniless, and
 if Mia has money, then:
 if Mia is proud of Mia's money, Sue is proud of Sue's money.

Corroboration for (51)–(52) comes from (53)–(54). In (53), A can be a felicitous response to B: presumably, A contests the presupposition of the $\forall$ -over-*if* reading of B. A' is an odd response, because neither reading presupposes *Mia has money*. The contradictory A'' shows that *If Mia has money, Sue has money* follows from what the $\forall$ -over-*if* reading of B presupposes, not from what it asserts. That A in (53) is indeed a response to the $\forall$ -over-*if* reading of B – rather than to its *if-over- $\exists$* reading – is shown by the fact that A in (54) is an odd response to the ellipsis-free B°, which has only the *if-over- $\exists$* reading (as we saw in 3.1).

- (53) B: If Mia is (either) penniless or proud of her money, then Sue is.
 A: I can neither agree nor disagree with you, because I think that Mia has money and Sue does not.
 A': #I can neither agree nor disagree with you, because I think that Mia has money.

- A": $\neg$ I can neither agree nor disagree with you, because I think that both Mia and Sue have money, Mia is proud of her money, and Sue is ashamed of hers.
- (54) B°: If Mia is (either) penniless or proud of her money, then Sue is (either) penniless or proud of her money.
- A: #I can neither agree nor disagree with you, because I think that Mia has money and Sue does not.

That what is contested by A in (53) – namely, *If Mia has money, Sue has money* – follows from what is presupposed by B, rather than from what B asserts, is consistent with (55), where a variation on the *Hey, wait a minute!* test from von Fintel (2004) is applied, and *that* is anaphoric to the assertion of B: only A* is a felicitous response to B, because only A* contests something that B presupposes.

- (55) B: If Mia is (either) penniless or proud of her money, then Sue is.
- A*: Hey, wait a minute! Before we even talk about that, I didn't know Mia has money only if Sue does!
- A": #Hey, wait a minute! Before we even talk about that, I didn't know Mia has money!
- A": #Hey, wait a minute! Before we even talk about that, I didn't know that!

Crucially, no K^{++} -based LF produces the $\forall$ -over-*if* reading of (48). In particular, (56) fails to produce that reading. The possible interpretations of prop_E in (56) are as in (57) (cf. (44)), on the assumption that *penniless* and *proud of her money* have the meanings in (58).

- (56) $\text{if}^{K^{++}} \wedge [\text{Mia} [\text{or}^{K^{++}} \wedge \text{penniless} \wedge [\text{proud of her money}]]] \wedge [\text{Sue} \vee \text{prop}_E]$
- (57) a. prop_E denotes $\text{f1prop}_E = \llbracket \wedge [\text{or}^{K^{++}} \wedge \text{penniless} \wedge [\text{proud of her money}]] \rrbracket^{w,c}$
- b. prop_E denotes f2prop_E , which is such that for all w :
- if $\llbracket \text{penniless} \rrbracket^{w,c}(\text{Mia}) = \text{True}$, $\text{f2prop}_E(w) = \llbracket \text{penniless} \rrbracket^{w,c}$,
- if $\llbracket \text{proud of her money} \rrbracket^{w,c}(\text{Mia}) = \text{True}$, $\text{f2prop}_E(w) = \llbracket \text{proud of her money} \rrbracket^{w,c}$.
- (58) a. $\llbracket \text{penniless} \rrbracket^{w,c} = [\lambda x. x \text{ has no money in } w]$
- b. $\llbracket \text{proud of her money} \rrbracket^{w,c} = [\lambda x. x \text{ has money in } w. x \text{ is proud of } x\text{'s money in } w]$

Paired with f1prop_E , (56) yields the meaning in (59). Paired with f2prop_E , (56) yields the meaning in (60). (59) accounts for the presupposition-free *if*-over- $\exists$ reading, but

(60) does not account for the $\forall$ -over-*if* reading, because that reading also comes out presupposition-free.

- (59) Pairing (56) with $f1prop_E$ (*if*-over- $\exists$; see (51))
 Presupposition of (56): –
 Assertion of (56): For all $w' \in EP_w$ such that Mia is penniless or proud of Mia's money in w' , Sue is penniless or proud of Sue's money in w' .
- (60) Pairing (56) with $f2prop_E$ (not $\forall$ -over-*if*; see (52))
 Presupposition of (56): –
 Assertion of (56): For all $w' \in EP_w$ such that Mia is penniless in w' , Sue is penniless in w' , and for all $w' \in EP_w$ such that Mia is proud of Mia's money in w' , Sue is proud of Sue's money in w' .

Let us, then, replace if^{K++} in (37) with $if^?$ in (61), which has an additional presupposition. For now, we leave ' $??$ ' unspecified.

- (61) $\llbracket if^? \rrbracket^{w,c}(p_1)$ is defined iff $p_1(w') \in \{\text{True}, \text{False}\}$.
 If defined, $\llbracket if^? \rrbracket^{w,c}(p_1) =$
 $\llbracket \lambda p_2: ?? \text{ or } p_2(w) \in \{\text{True}, \text{False}\}. \{w' \mid w' \in EP_w \text{ and } p_1(w') = \text{True}\} \subseteq \{w' \mid p_2(w') = \text{True}\} \rrbracket$.

Suppose, for the sake of the discussion, that whatever ' $??$ ' is, the *if*-over- $\exists$ meaning of (48) described in (51) is derived when (62) is paired with $f1prop_E$, and the $\forall$ -over-*if* meaning described in (52) is derived when (62) is paired with $f2prop_E$.⁸

- (62) $if^? \wedge [Mia \text{ } or^{K++} \wedge penniless \wedge [proud \text{ of her money}]] \wedge [Sue \vee prop_E]$

Crucially, this cannot – in principle – account for the fact that both (63a) (= (48)) and (63b) have both the *if*-over- $\exists$ meaning described in (51) and the $\forall$ -over-*if* meaning described in (52).

- (63) a. If Mia is penniless or proud of her money, then Sue is.
 b. If Mia is proud of her money or penniless, then Sue is.

The reason is this. On the one hand, (62) and (64) – which are accommodation free – are not equivalent to each other, on any of the interpretive options of $prop_E$. While (62) does not presuppose *Mia has money* (by assumption), regardless of how

⁸ Notice that if ' $??$ ' = ' $p_1(w) = \text{False}$ ' (as in Karttunen (1974)), we predict the $\forall$ -over-*if* presupposition of (48) to be *If Mia is proud of her money, Sue has money*. However, the attested $\forall$ -over-*if* presupposition of (48) described in (52a) is *If Mia has money, Sue has money*. We come back to this issue in Section 4.

prop_E is interpreted, (64) presupposes *Mia has money*, regardless of how prop_E is interpreted, because both $\text{if}^?$ and or^{K++} “pass up” the presuppositions of their leftmost argument.

$$(64) \quad \text{if}^? \wedge [\text{Mia } [\text{or}^{K++} \wedge [\text{proud of her money}] \wedge \text{penniless}]] \wedge [\text{Sue } \vee \text{prop}_E]$$

On the other hand, some accommodation-enriched pairs of LFs of (63a,b) do come out equivalent, but the $\forall$ -over-*if* reading of (63b) is not derived. For example, (62) and (65) are equivalent only when paired with f1prop_E , which only yields the *if*-over- $\exists$ reading (notice that in (65), the second interpretive option of prop_E is f3prop_E , which is such that for all w , if Mia is proud of her money in w , $\text{f3prop}_E(w) = [\lambda x. x \text{ has money in } w \text{ and } x \text{ is proud of } x\text{'s money in } w]$). (66a) and (66b) are always equivalent, but $\wedge [\text{Mia has money}]$ is globally accommodated. In short, there is no way to derive the $\forall$ -over-*if* reading of (63b) within $K++$.⁹

$$(65) \quad \text{if}^? \wedge [\text{Mia } [\text{or}^{K++} \wedge [\text{and}^{K++} \wedge [\text{has money}] \wedge [\text{proud of her money}]] \wedge \text{penniless}]] \wedge [\text{Sue } \vee \text{prop}_E]$$

$$(66) \quad \begin{array}{ll} \text{a.} & \text{and}^{K++} \wedge [\text{Mia has money}] \wedge [\text{if}^? \wedge [\text{Mia } [\text{or}^{K++} \wedge \text{penniless} \wedge [\text{proud of her money}]]] \wedge [\text{Sue } \vee \text{prop}_E]] \\ \text{b.} & \text{and}^{K++} \wedge [\text{Mia has money}] \wedge [\text{if}^? \wedge [\text{Mia } [\text{or}^{K++} \wedge [\text{proud of her money}]] \wedge \text{penniless}]] \wedge [\text{Sue } \vee \text{prop}_E] \end{array}$$

Let us summarize. On the one hand, an asymmetric *or* – with or without accommodation – fails to account for the equivalence of *If x is P or Q , then y is* and *If x is Q or P , then y is*. On the other hand, if we merely concede that *or* is inherently symmetric, we will predict *If x is P or Q , then y is* and *If x is Q or P , then y is* to be equivalent, but we will not account for their $\forall$ -over-*if* reading, unless we figure out what ‘___?’ in (61) is. Sections 4–5 are concerned with that.

9 Assuming, alternatively, that the syntax somehow generates for (63a,b) the LFs in (i) – where two “small” plain conditional LFs are conjoined – would not solve the problem, as at least (ib) presupposes that Mia has money. One might argue that the f2 option is available for (65), but f2 will not work for (30) and its counterpart once we acknowledge that their disjuncts are presuppositional.

(i) a. $\text{and}^{K++} \wedge [\text{if}^? \wedge [\text{Mia penniless}] \wedge [\text{Sue penniless}]] \wedge [\text{if}^? \wedge [\text{Mia proud of her money}] \wedge [\text{Sue proud of her money}]]$
 b. $\text{and}^{K++} \wedge [\text{if}^? \wedge [\text{Mia proud of her money}] \wedge [\text{Sue proud of her money}]] \wedge [\text{if}^? \wedge [\text{Mia penniless}] \wedge [\text{Sue penniless}]]$

4 An attempt to spell out the second presupposition of *if*

In 4.1 we discuss in detail what the additional presupposition in (61) should capture, and in 4.2 we explore a failed attempt to spell it out. Our own proposal is presented in Section 5.

4.1 The context-dependency of filtering

To figure out what ‘___? ___’ in (61) stands for, we first note that (68) describes the $\forall$ -over-*if* reading of the Rooth-Partee conditional (67) more accurately (compared to (52)).

(67) If Mia is penniless or proud of her money, then Sue is.

(68) Presupposition of the $\forall$ -over-*if* reading of (67):

If Mia has money, the presupposition of (69b) is true.

Assertion of the $\forall$ -over-*if* reading of (67):

The assertion of (69a) is true, and if Mia has money, the assertion of (69b) is true.

(69) a. If Mia is penniless, Sue is penniless.

b. If Mia is proud of her money, Sue is proud of her money.

To figure out what (67) presupposes, given (68), we have to figure out what (69b) presupposes. As first observed in Karttunen (1973), when p_1 in *if* $p_1 p_2$ does not entail all the presuppositions of p_2 , the non-entailed presuppositions of p_2 need not be filtered out. (69b), where *Sue has money* is not entailed by *Mia is proud of her money* illustrates this: indeed, (70) is a coherent exchange, suggesting that *Sue has money* need not be filtered out.

(70) B: If Mia is proud of her money, Sue is proud of her money.

A: I can neither agree nor disagree with you, because I think Sue has no money.

In addition, (71) is a contradictory response to (69b); this confirms that what is contested by A in (70) – namely, *Sue has money* – follows from what (69b) presupposes, not from what it asserts. The *Hey, wait a minute!* test, applied in (72), is consistent with that.

(71) $\perp$ I can neither agree nor disagree with you, because I think that Mia is proud of her money, but Sue is ashamed of hers.

- (72) B: If Mia is proud of her money, Sue is proud of her money.
 A: Hey, wait a minute! Before we even talk about that, I didn't know Sue has money!
 A': #Hey, wait a minute! Before we even talk about that, I didn't know Mia is proud of her money!
 A'': #Hey, wait a minute! Before we even talk about that, I didn't know that!

However, as (73) shows, *Sue has money* is filtered out in some contexts.

- (73) I don't know if Sue has any money, but I do know that the dynamics between Mia and Sue makes it impossible for Mia to be proud of her money when Sue is penniless. Furthermore, I'm pretty sure that if Mia is proud of her money, then Sue is.

This means that the additional presupposition in (61) should predict, not that (69b) presupposes *Sue has money*, but that it may presuppose *Sue has money*, depending on the context. Given (68), the additional presupposition in (61) should predict, not that the $\forall$ -over-if reading (67) presupposes *If Mia has money, Sue has money*, but that it may presuppose *If Mia has money, Sue has money*, depending on the context.

When does (69b) presuppose *Sue has money*, and when does the $\forall$ -over-if reading of (67) presuppose *If Mia has money, Sue has money*? Before answering this question, let us look more closely at the presuppositions of plain conditionals and their corresponding conjunctions and disjunctions. Consider (74)–(76), which illustrate familiar redundancy contrasts as well as the following fact: filtering is context-dependent, not only in conditionals (as (70) and (73) already show), but also in conjunctions and disjunctions (see Karttunen (1973); see also Geurts (1996, 1999), Mandelkern (2016), Winter (2025) and others).

- (74) a. If Mia has money, her staff take care of her money.
 b. #If Mia's staff take care of her money, she has money.
 (75) a. Mia has money, and her staff take care of her money.
 b. #Mia's staff take care of her money, and she has money.
 (76) a. (Either) Mia has no money, or her staff take care of her money.
 b. (Either) Mia's staff take care of her money, or she has no money.

Mia has a staff is a presupposition of *Mia's staff take care of her money* that is not entailed by *Mia has money*. In (77), but not in (78), A can be a felicitous response to any variant of B. This suggests that (i), (ii), (iii) and (iv) may inherit the presupposition *Mia has a staff* from *Mia's staff take care of her money* in (77), where they are uttered out of the blue, but not in (78), where they are uttered after establishing the truth of the law-like proposition *Mia cannot have money unless she has a staff*.

- (77) B: (i) If Mia has money, her staff take care of her money.
 (ii) Mia has money and her staff take care of her money.
 (iii) Either Mia has no money, or her staff take care of her money.
 (iv) Either Mia's staff take care of her money, or she has no money.
 A: I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think that Mia doesn't have a staff.
- (78) B: I don't know if Mia has a staff, but I do know that Mia's lifestyle and social status require that if she has money, she has a staff, and I'm also pretty sure that
 (i) if Mia has money, her staff take care of her money.
 (ii) Mia has money, and her staff take care of her money.
 (iii) either Mia has no money, or her staff take care of her money.
 (iv) either Mia's staff take care of her money, or she has no money.
 A: #I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think that Mia doesn't have a staff.
 A': I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think that nothing prevents Mia from having money but no staff.

Moreover, in (79) – where B explicitly denies the law-like proposition *Mia cannot have money unless she has a staff* – (i)-(iv) can, again, inherit the presupposition *Mia has a staff*.

- (79) B: I know that Mia's lifestyle and social status do not prevent her from having money but no staff, but I'm pretty sure that
 (i) if Mia has money, her staff take care of her money.
 (ii) Mia has money, and her staff take care of her money.
 (iii) either Mia has no money, or her staff take care of her money.
 (iv) either Mia's staff take care of her money, or she has no money.
 A: I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think that Mia doesn't have a staff.
 A': #I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think nothing prevents Mia from having money but no staff.

In addition, (80) is a contradictory response to B in (77) and (79). This confirms that what is contested by A in (77) and (79) – namely, *Mia has a staff* – does not follow from what is asserted by (i), (ii), (iii) or (iv).

- (80) ⊥I can neither agree nor disagree with your claim that (i)/(ii)/(iii)/(iv), because I think that Mia has money but her staff don't take care of her money.

Law-like presuppositions are pragmatic presuppositions; they are determined by what is established in the discourse, not by the lexical meanings of the connectives. And so, we say that *If Mia has money, her staff take care of her money* semantically presupposes *Mia has a staff* only if the law-like *Mia can't have money unless she has a staff* is not (pragmatically) presupposed. This is further reflected by the contrast between (81) and (82) (see Pearson [2010]). Similar contrasts arise with the corresponding conjunctions and disjunctions.

- (81) C: If Mia has money, her staff take care of her money.
 D: Hey, wait a minute! Before we even talk about that, I didn't know Mia has a staff!
 C: She does, so yes, I insist that if she has money, her staff take care of her money.
- (82) E: If Mia has money, her staff take care of her money.
 D: Hey, wait a minute! Before we even talk about that, I didn't know Mia has a staff!
 E: Well, I don't know if she does, but in her milieu it's impossible to have money but no staff, so yes, I insist that if she has money, her staff take care of her money.

A naïve attempt to state the empirical generalizations regarding what *if*, *and* and *or* s-presuppose (i.e., semantically presuppose) is given in (83)–(84). As before, EP_w is the set of worlds compatible with the relevant body of knowledge in w .

- (83) For all q in $\{if\ p_1\ p_2, p_1\ and\ p_2\}$, q s-presupposes in w (a)-(b) and nothing else:
 (a) every p s-presupposed by p_1 in w , and
 (b) every p s-presupposed by p_2 in w such that: (i) p is not entailed by p_1 , and (ii) the law-like propositions that are (pragmatically) presupposed in w allow p_1 to be true while p is false (and so, in at least one world in EP_w , p_1 is true and p is false).
- (84) $p_1\ or\ p_2$ s-presupposes in w (a)-(b) and nothing else:
 (a) every p s-presupposed by p_1 in w or every p s-presupposed by p_2 in w , and
 (b) every p s-presupposed by p_2 in w such that: (i) p is not entailed by p_1 , and (ii) the law-like propositions that are (pragmatically) presupposed in w allow p_1 to be true while p is false (and so, in at least one world in EP_w , p_1 is true and p is false),
 and
 every p s-presupposed by p_1 in w such that: (i) p is not entailed by p_2 , and (ii) the law-like propositions that are (pragmatically) presupposed in w allow p_2 to be true while p is false (and so, in at least one world in EP_w , p_2 is true and p is false).

The redundancy contrasts in (74)–(76) are indeed captured by (83a), (84a) and CP: both *#If Mia's staff take care of her money, she has money* and *#Mia's staff take care of her money and she has money* contain the uninformative *she has money*. However, the context dependency of filtering illustrated by (77)–(82) is clearly not captured by (83b) and (84b). Take (83b); since it allows the s-presuppositions of p_2 that are also s-presuppositions of *if* p_1 p_2 to be false in some world of EP_w , it obviously fails to require those s-presuppositions to be true in all the worlds of EP_w . To remedy this, we distinguish between classical entailment ($\Rightarrow$), which has the standard definition in (85a), and the $\sim$ relation defined in (85b), where LAW_w is the set of worlds compatible with the laws that are in effect in w (thus, for example, although $\neg(\text{Mia has money} \Rightarrow \text{Mia has a staff})$, there are worlds w such that *Mia has money* $\sim_w$ *Mia has a staff*).¹⁰

- (85) a. $p \Rightarrow q$ iff for all w such that p is true in w , q is true in w .
 b. $p \sim_w q$ iff for all w' in LAW_w such that p is true in w' , q is true in w' .

Both the redundancy contrasts in (74)–(76) and the context-dependency of filtering illustrated by (77)–(82) are captured by (86)–(87).

- (86) For all q in $\{\text{if}^{\text{plain}} p_1 p_2, p_1 \text{ and } p_2\}$, q s-presupposes in w (a)-(b) and nothing else:
 (a) every p such that p_1 s-presupposes p in w ,
 (b) every p in S_w , where:
 $S_w = \{p \mid p_2 \text{ s-presupposes } p \text{ in } w, \text{ and } EP_w \subseteq \{w' \mid \neg(p_1 \sim_{w'} p)\}, \text{ and}$
 $\text{for all } r \text{ such that } p \Rightarrow r: \neg(p_1 \Rightarrow r) \text{ or } p_1 \text{ s-presupposes } r \text{ in } w\}$
 (i.e., for all q in $\{\text{if}^{\text{plain}} p_1 p_2, p_1 \text{ and } p_2\}$, q has a truth value in w iff for all p such that p_1 s-presupposes p in w or $p \in S_w$, $EP_w \subseteq \{w' \mid p \text{ is true in } w'\}$).
- (87) p_1 or p_2 s-presupposes in w (a)-(b) and nothing else:
 (a) every p such that p_1 s-presupposes p in w or
 every p such that p_2 s-presupposes p in w ,
 (b) every p in $S_w^* \cup S_w^\circ$, where:
 $S_w^* = \{p \mid p_2 \text{ s-presupposes } p \text{ in } w, \text{ and } EP_w \subseteq \{w' \mid \neg(\text{neg } p_1 \sim_{w'} p)\}, \text{ and}$
 $\text{for all } r \text{ such that } p \Rightarrow r: \neg(\text{neg } p_1 \Rightarrow r) \text{ or } p_1 \text{ s-presupposes } r \text{ in } w\}, \text{ and}$
 $S_w^\circ = \{p \mid p_1 \text{ s-presupposes } p \text{ in } w, \text{ and } EP_w \subseteq \{w' \mid \neg(\text{neg } p_2 \sim_{w'} p)\}, \text{ and}$
 $\text{for all } r \text{ such that } p \Rightarrow r: \neg(\text{neg } p_2 \Rightarrow r) \text{ or } p_2 \text{ s-presupposes } r \text{ in } w\}$
 (i.e., p_1 or p_2 has a truth value in w iff: for some $p \in \{\text{the strongest s-presupposition in } w \text{ of } p_1, \text{ the strongest s-presupposition in } w \text{ of } p_2\}$, $EP_w \subseteq \{w' \mid p \text{ is true in } w'\}$, and for all $p \in S_w^* \cup S_w^\circ$, $EP_w \subseteq \{w' \mid p \text{ is true in } w'\}$).

¹⁰ Since we treat the concept of a law as a primitive, we do not provide a definition for it (but we assume that laws come in various flavors: a law can be authority-based, habit-based, disposition-based, custom-based, and so on).

To illustrate, consider *If Mia has money, her staff take care of her money*, where $p_1 = \text{Mia has money}$ and $p_2 = \text{Mia's staff take care of her money}$. For any w , the s-presupposition of p_2 is *Mia has money and a staff*, which entails both p_1 and *Mia has a staff*. By (86b), p_1 is not in S_w because p_1 does not s-presuppose p_1 in any world. Indeed, p_1 is filtered out (i.e., it need not be true in all the worlds of EP_w for *If Mia has money, her staff take care of her money* to be uttered in w). *Mia has money and a staff* is not in S_w either (for essentially the same reason). On the other hand, whether *Mia has a staff* is in S_w depends on the laws that are in effect in the worlds of EP_w . If *Mia has money* $\sim_{w'} \text{Mia has a staff}$ in some w' of EP_w (because, say, this is a requirement in Mia's social milieu in w'), *Mia has a staff* is not in S_w (i.e., need not be true in the worlds of EP_w). This is the case in (78). If $\neg(\text{Mia has money} \sim_{w'} \text{Mia has a staff})$ holds in all w' of EP_w , *Mia has a staff* is in S_w (i.e., true in the worlds of EP_w). This is the case in (79); it stands to reason that it is also the case in (77).

Now consider Rooth-Partee conditionals. In Section 3 we established the following: (a) that the *if-over- $\exists$* reading of *If x is P or Q , then y is* is the same as the only reading of the plain conditional *If x is P or Q , then y is P or Q* , and (b) that the *$\forall$ -over-if* reading of *If x is P or Q , then y is* expresses the conjunction of two plain conditionals, namely: '*if x is P has a truth value, then *If x is P , y is P is true*' and '*if x is Q has a truth value, then *If x is Q , y is Q is true*'. It is reasonable, then, to hypothesize that the s-presuppositions of the *if-over- $\exists$* and *$\forall$ -over-if* readings of *If x is P or Q , then y is* are as described in (88)–(89).**

- (88) In w , the *if-over- $\exists$* reading of *If x is P or Q , then y is* s-presupposes the following and nothing else:
 every q such that, by (86), $\text{if}^{\text{plain}} [x \text{ is } P \text{ or } x \text{ is } Q] [y \text{ is } P \text{ or } y \text{ is } Q]$ s-presupposes q in w .
- (89) In w , the *$\forall$ -over-if* reading of *If x is P or Q , then y is* s-presupposes a-c. and nothing else:
- every q such that, by (87), $x \text{ is } P \text{ or } x \text{ is } Q$ s-presupposes q in w ,
 - $[\lambda w'. \text{if } x \text{ is } P \text{ has a truth value in } w', \text{ then for every } q \text{ such that, by (86), } \text{if}^{\text{plain}} [x \text{ is } P] [y \text{ is } P] \text{ s-presupposes } q \text{ in } w': q \text{ is true in } w']$,
 - $[\lambda w'. \text{if } x \text{ is } Q \text{ has a truth value in } w', \text{ then for every } q \text{ such that, by (86), } \text{if}^{\text{plain}} [x \text{ is } Q] [y \text{ is } Q] \text{ s-presupposes } q \text{ in } w': q \text{ is true in } w']$.

According to (88)–(89), the *if-over- $\exists$* reading of (90a) and (90b) is s-presupposition-free, and their *$\forall$ -over-if* reading has the s-presupposition in (91).

- (90) a. If Mia is penniless or proud of her money, then Sue is.
 b. If Mia is proud of her money or penniless, then Sue is.

- (91) In w, the $\forall$ -over-if reading of (90a) and (90b) s-presupposes:
 if Mia has money in w, and the laws that are (pragmatically) presupposed in w allow Mia to be proud of her money while the s-presuppositions in w of *Sue is proud of her money* are false, then the s-presupposition in w of *Sue is proud of her money* is true
 (i.e., if Mia has money and the laws allow Mia to be proud of her money while Sue has no money, Sue has money.)

Indeed, the exchanges in (92)–(94), and the fact that the same judgments are elicited if (90a) – which is embedded in these exchanges – is replaced with (90b), confirm (91).

- (92) B: If Mia is penniless or proud of her money, then Sue is.
 A: I can neither agree nor disagree with you, because I don't think that Mia has money only if Sue does.
- (93) B: I don't know if the dynamics between Mia and Sue makes it impossible for Mia to have money while Sue is penniless, but I do know that the dynamics between them makes it impossible for Mia to have money and be proud of it while Sue is penniless. Moreover, I know that if Mia is penniless or proud of her money, then Sue is.
 A: #I can neither agree nor disagree with your claim that if Mia is penniless or proud of her money, then Sue is, because I think that Mia has money and she is ashamed of her money, but Sue is penniless.
 A': I can neither agree nor disagree with your claim that if Mia is penniless or proud of her money, then Sue is, because I think that Mia has money and she is proud of her money, but Sue is penniless.
- (94) B: I think that the dynamics between Mia and Sue is such that Mia can have money and be proud of it while Sue is penniless. Moreover, I'm pretty sure that if Mia is penniless or proud of her money, then Sue is.
 A: I can neither agree nor disagree with your claim that if Mia is penniless or proud of her money, then Sue is, because I think that Mia has money and she is ashamed of her money, but Sue is penniless.
 A': #I can neither agree nor disagree with your claim that if Mia is penniless or proud of her money, then Sue is, because I think that Mia has money and she is proud of her money, but Sue is penniless.

That the $\forall$ -over-if reading of (90a) can intuitively presuppose *If Mia has money, Sue has money*, provided that according to the (pragmatically) presupposed laws, Mia can have money and be proud of it while Sue is penniless, is consistent with (95)–(96) (cf. (81)–(82)). The same judgments are elicited if (90a) – which is embedded in (95)–(96) – is replaced with (90b).

- (95) C: If Mia is penniless or proud of her money, then Sue is.
 D: Hey, wait a minute! Before we even talk about that, I didn't know that Mia can't have money unless Sue does!
 C: That's indeed the case. So yes, I insist that if Mia is penniless or proud of her money, then Sue is.
- (96) E: If Mia is penniless or proud of her money, then Sue is.
 D: Hey, wait a minute! Before we even talk about that, I didn't know that Mia can't have money unless Sue does!
 E: Well, I don't know if that's the case, but I do know that the dynamics between Mia and Sue is such that Mia can't have money and be proud of her money unless Sue has money, so I insist that if Mia is penniless or proud of her money, then Sue is.

That only the $\forall$ -over-*if* reading is relevant in (92)–(94) is confirmed by (97), where B° is ellipsis-free, and has only the *if*-over- $\exists$ reading.

- (97) B°: If Mia is penniless or proud of her money/proud of her money or penniless, then Sue is penniless or proud of her money/proud of her money or penniless.
 A: #I can neither agree nor disagree with you, because I don't think that Mia has money only if Sue does.

That (98) is a contradictory response to B in (92)–(94) confirms that what is contested by A in (92) and (94) – *If Mia has money, Sue has money* – is not asserted by B.

- (98) ⊥I can neither agree nor disagree with your claim that if Mia is penniless or proud of her money, then Sue is, because I think that Mia and Sue have money, Mia is proud of her money but Sue is ashamed of hers.

The same tests will show that the intuitive s-presuppositions, expected given (86)–(87) and (88)–(89), also arise regarding (99), where *not happy* does not entail *has money*: both the *if*-over- $\exists$ and $\forall$ -over-*if* readings of (99a,b) have the s-presupposition – projected from the disjunctive antecedent – that Mia has money, provided that the (pragmatically) presupposed laws allow Mia to be unhappy and have no money. Their $\forall$ -over-*if* reading s-presupposes *If Mia has money, Sue has money*, provided that the (pragmatically) presupposed laws allow Mia to have money and be proud of her money while Sue has no money.

- (99) a. If Mia is (either) happy or proud of her money, then Sue is.
 b. If Mia is (either) proud of her money or happy, then Sue is.

In 4.2 we discuss a failed attempt to derive (86)–(87) and (88)–(89). The attempt fails because it cannot account for $\forall$ -over-*if* readings of Rooth-Partee conditionals.

4.2 Rooth-Partee conditionals and the Proviso Problem

In Karttunen (1974), the extra presupposition in (61) is ‘if p_1 is true, p_2 has a truth value’. We established in Section 3 that *or* is symmetric. In Karttunen and Peters (1979), the connectives have the same meanings they have in Karttunen (1974), except that *or* is symmetric. K/P , spelled out below, incorporates both these ideas: $\mathbf{if}^{K/P}$ and $\mathbf{and}^{K/P}$ are asymmetric, $\mathbf{or}^{K/P}$ is symmetric, and all three have Karttunen-style conditional presuppositions.

- (100) $\llbracket \mathbf{if}^{K/P} \rrbracket^{w,c}(p_1)$ is defined iff $p_1(w) \in \{\text{True}, \text{False}\}$.
 If defined, $\llbracket \mathbf{if}^{K/P} \rrbracket^{w,c}(p_1) =$
 $[\lambda p_2: p_1(w) = \text{False} \text{ or } p_2(w) \in \{\text{True}, \text{False}\}. \{w' \mid w' \in \text{EP}_w \text{ and } p_1(w') = \text{True}\} \subseteq \{w' \mid p_2(w') = \text{True}\}].$
- (101) When P_1 is a proposition, $\llbracket \mathbf{and}^{K/P} \rrbracket^{w,c}(P_1)$ is defined iff $\llbracket \mathbf{if}^{K/P} \rrbracket^{w,c}(P_1)$ is defined.
 If defined, $\llbracket \mathbf{and}^{K/P} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2: P_1(w) = \text{False} \text{ or } P_2(w) \in \{\text{True}, \text{False}\}. P_1(w) = P_2(w) = \text{True}].$
 When P_1 is a property of individuals, $\llbracket \mathbf{and}^{K/P} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2. [\lambda x: \llbracket \mathbf{and}^{K/P} \rrbracket^{w,c}(P_1^x)(P_2^x) \in \{\text{True}, \text{False}\}. \llbracket \mathbf{and}^{K/P} \rrbracket^{w,c}(P_1^x)(P_2^x)]]].$
- (102) When P_1 is a proposition, $\llbracket \mathbf{or}^{K/P} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2: (P_1(w) = \text{True} \text{ or } P_2(w) \in \{\text{True}, \text{False}\}) \text{ and}$
 $(P_2(w) = \text{True} \text{ or } P_1(w) \in \{\text{True}, \text{False}\}).$
 $P_1(w) = \text{True} \text{ or } P_2(w) = \text{True}].$
 When P_1 is a property of individuals, $\llbracket \mathbf{or}^{K/P} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2. [\lambda x: \llbracket \mathbf{or}^{K/P} \rrbracket^{w,c}(P_1^x)(P_2^x) \in \{\text{True}, \text{False}\}. \llbracket \mathbf{or}^{K/P} \rrbracket^{w,c}(P_1^x)(P_2^x)]]].$
- (103) $Z^x = [\lambda w: Z(w^x)(x) \in \{\text{True}, \text{False}\}. Z(w^x)(x)]$

A well-known problem regarding the Karttunen-style conditional presupposition, dubbed “the Proviso Problem” in Geurts (1996, 1999), is that it is too weak to capture the fact that when p_2 s-presupposes q but $\neg(p_1 \Rightarrow q)$, it is often the case that *if* p_1 p_2 intuitively presupposes q , not merely ‘if p is true, q has a truth value’ (the same problem arises regarding *and* and *or*). Indeed, as we saw, (104) often intuitively presupposes *Mia has a staff*.

- (104) If Mia has money, her staff take care of her money.

Since here we describe the s-presuppositions of $\mathbf{if}^{\text{plain}}$ as in (86) – the relevant part of which is reproduced in (105) – we state the Proviso Problem slightly differently, as follows: LF (106) predicts – contra (105)+CP – that (104) is potentially felicitous in w_c

even if not every element of S_{w_c} is true in all the worlds of EP_{w_c} . This is shown in (107), where Mia has no staff in at least one world of EP_{w_c} , yet (107iii) holds.

- (105) $if^{plain} p_1 p_2$ s-presupposes in w (at least) every element of S_w , where:
 $S_w = \{p \mid p_2 \text{ s-presupposes } p \text{ in } w, \text{ and } EP_w \subseteq \{w' \mid \neg(p_1 \rightsquigarrow_w p)\}\}$, and
 for all r such that $p \Rightarrow r: \neg(p_1 \Rightarrow r) \text{ or } p_1 \text{ s-presupposes } r \text{ in } w\}$.
- (106) $if^{K/P} \wedge [Mia \text{ has money}] \wedge [Mia's \text{ staff take care of Mia's money}]$
- (107) LF (106) is potentially CP-compliant relative to c when:
- (i) $\{w \mid w \in EP_{w_c} \text{ and Mia has money in } w\} \neq \emptyset$, and
 $\{w \mid w \in EP_{w_c} \text{ and Mia has money in } w\} \subseteq \{w \mid \text{Mia has a staff in } w\}$,
 - (ii) $\{w \mid w \in EP_{w_c} \text{ and Mia has no staff in } w\} \neq \emptyset$, and
 - (iii) $\{w \mid w \in EP_{w_c} \text{ and } \neg(Mia \text{ has money} \rightsquigarrow_w Mia \text{ has a staff})\} = EP_{w_c}$.

In Section 3 we rejected accommodation as a means to explain the intuitive symmetry of *or*. Since the Proviso Problem and the Intuitive-symmetry-of-*or* Problem are distinct problems, we should still ask whether accommodation may solve the former (even if it cannot solve the latter). Suppose, then, that (104) has – in addition to (106) – the LF in (108a), where $\wedge[Mia \text{ has a staff}]$ is accommodated globally, and the LF in (108b), where $\wedge[Mia \text{ has a staff}]$ is accommodated non-globally. Suppose also, as is commonly assumed, that CP not only bans uninformativity, but also favors maximal informativity.

- (108) a. $and^{K/P} \wedge [Mia \text{ has a staff}] \wedge [if^{K/P} \wedge [Mia \text{ has money}] \wedge [Mia's \text{ staff take care of Mia's money}]]$
 b. $if^{K/P} \wedge [Mia \text{ has money}] \wedge [and^{K/P} \wedge [Mia \text{ has a staff}] \wedge [Mia's \text{ staff take care of Mia's money}]]$

Since (108a) is stronger than both (106) and (108b), it is deemed by the hearer of (104) as the intended LF when she is agnostic about *Mia has a staff*. When the discourse participants in c already agree that Mia has a staff, $\wedge[Mia \text{ has a staff}]$ is uninformative in both (108a) and (108b) relative to c . Either way, the Proviso Problem is circumvented. Similar reasoning also guarantees that (109a) and (109b) (like (25a) and (26a)) are intuitively equivalent.

- (109) a. (Either) Mia has no money or her staff take care of her money.
 b. (Either) Mia's staff take care of her money or she has no money.
- (110) a. $and^{K/P} \wedge [Mia \text{ has a staff}] \wedge [or^{K/P} \wedge [Mia \text{ has no money}] \wedge [Mia's \text{ staff take care of Mia's money}]]$ (cf. (25b))
 b. $and^{K/P} \wedge [Mia \text{ has a staff}] \wedge [or^{K/P} \wedge [and^{K/P} \wedge [Mia \text{ has money}] \wedge [Mia's \text{ staff take care of Mia's money}]] \wedge [Mia \text{ has no money}]]$ (cf. (26b))

Even if we concede that this reasoning achieves the desired results regarding (104) and (109), we are forced to acknowledge that it cannot yield the desired result regarding the Rooth-Partee conditional (111), which has a $\forall$ -over-*if* reading. The relevant part of the schematic $\forall$ -over-*if* s-presupposition, according to (89), is reproduced in (112).

(111) If Mia is penniless or proud of her money, then Sue is.

(112) In w , the $\forall$ -over-*if* reading of *If x is P or Q , then y is* s-presupposes at least the following:

$[\lambda w'. \text{if } x \text{ is } P \text{ has a truth value in } w', \text{ then for every } q \text{ such that, by (86)/(105), } if^{\text{plain}} [x \text{ is } P] [y \text{ is } P] \text{ s-presupposes } q \text{ in } w': q \text{ is true in } w'], \text{ and}$

$[\lambda w'. \text{if } x \text{ is } Q \text{ has a truth value in } w', \text{ then for every } q \text{ such that, by (86)/(105), } if^{\text{plain}} [x \text{ is } Q] [y \text{ is } Q] \text{ s-presupposes } q \text{ in } w': q \text{ is true in } w']$.

When (113) is paired with $f2\text{prop}_E$, (114)–(115) hold and the resulting s-presupposition is this: if Mia has money and is proud of her money, Sue has money. The Proviso Problem rears its head, because – contra (112)+CP – (116) holds.

(113) $if^{K/P} \wedge [Mia [or^{K/P} \wedge penniless \wedge [proud of her money]]] \wedge [Sue \vee \text{prop}_E]$

(114) if $\llbracket penniless \rrbracket^{w,c}(Mia) = \text{True}$, $f2\text{prop}_E(w) = \llbracket penniless \rrbracket^{w,c}$,
if $\llbracket proud of her money \rrbracket^{w,c}(Mia) = \text{True}$, $f2\text{prop}_E(w) = \llbracket proud of her money \rrbracket^{w,c}$.

(115) a. $\llbracket penniless \rrbracket^{w,c} = [\lambda x. x \text{ has no money in } w]$

b. $\llbracket proud of her money \rrbracket^{w,c} = [\lambda x: x \text{ has money in } w. x \text{ is proud of } x\text{'s money in } w]$

(116) If paired with $f2\text{prop}_E$, (113) is potentially CP-compliant relative to c when:

(i) $\{w \mid w \in EP_{w_c} \text{ and Mia has money and is proud of Mia's money in } w\} \neq \emptyset$, and

$\{w \mid w \in EP_{w_c} \text{ and Mia has money and is proud of Mia's money in } w\} \subseteq \{w \mid \text{Sue has money in } w\}$,

(ii) $\{w \mid w \in EP_{w_c} \text{ and Mia has money and Sue has no money}\} \neq \emptyset$, and

(iii) $\{w \mid w \in EP_{w_c} \text{ and } \neg(Mia \text{ is proud of her money } \sim_w \text{ Sue has money})\} = EP_{w_c}$.

LF (117) – where $\wedge[Mia \text{ has money}]$ is globally accommodated – does not circumvent the Proviso Problem. (117) is stronger than (113), but it does not yield the intuitive $\forall$ -over-*if* presupposition: the accommodated proposition in (117) – *Mia has money* – is not the s-presupposition expected given (112).

- (117) $\text{and}^{K/P} \wedge [\text{Mia has money}] \wedge [\text{if}^{K/P} \wedge [\text{Mia } [\text{or}^{K/P} \wedge \text{penniless} \wedge [\text{proud of her money}]]] \wedge [\text{Sue} \vee \text{prop}_E]]$

In fact, no K/P-based LF of (111) circumvents the Proviso Problem. We conclude that while accommodation+CP might circumvent the Proviso Problem in plain conditionals, it fails to do so in Rooth-Partee conditionals. We add this conclusion to our conclusion from Section 3 (namely, that combining accommodation with a lexically asymmetric *or* fails to account for the intuitive equivalence of *If x is P or Q, then y is* and *If x is Q or P, then y is*). With these two conclusions, we turn to K/P*, where *or* is inherently symmetric – as in K/P – but the conditional presuppositions of the connectives differ radically from the conditional presuppositions in K/P.

5 The proposal

Our proposal relies on the theory of plain indicative conditionals in Heim (1992) (the precursor of which is Stalnaker [1975]), and its development in Pérez Carballo (2009). We extend Heim's analysis to disjunctions and Rooth-Partee conditionals.

The task of predicting the intuitive presuppositions of (118a) includes: (i) predicting that (118a) s-presupposes that Mia has a staff, provided that the pragmatically presupposed laws allow Mia to have money without having a staff (see 4.1); (ii) ensuring that (118a) does not s-presuppose everything that the factive (118b) s-presupposes (in particular, *If Mia has money, she has a staff*; see Geurts (1996, 1999) and 4.2); and (iii) predicting that on its 'de dicto' reading, (118c) s-presupposes that Fred believes what (118a) s-presupposes (in general, *x believes/knows p* intuitively presupposes that *x* believe what *p* intuitively presupposes; see Karttunen (1974)).

- (118) a. If Mia has money, her staff take care of her money.
 b. Fred knows that if Mia has money, she has a staff.
 c. Fred believes/knows that if Mia has money, her staff take care of her money.

To that end, we introduce $\text{if}^{K/P*}$ in (119), where RES_c is a similarity relation supplied by context *c*, such that ' $(w, (w', w'')) \in \text{RES}_c$ ' says: *w'* resembles *w* more than *w''* resembles *w*. The CLOS function is described in (120). The definedness condition of $\text{CLOS}_{w,X,\text{RES}_c}(Y)$ essentially does the work of the Limit Assumption (which, as claimed in Lewis (1973), must be adopted by any theory of *if* p_1 p_2 that makes reference to 'the closest p_1 -world(s)').

- (119) $\llbracket \text{if}^{K/P*} \rrbracket^{w,c}(p_1)$ is defined iff $p_1(w) \in \{\text{True}, \text{False}\}$.
 If defined, $\llbracket \text{if}^{K/P*} \rrbracket^{w,c}(p_1) =$
 $\lambda p_2: \text{CLOS}_{w,EP_w,\text{RES}_c}(\{w'' \mid p_1(w'') = \text{True}\}) \subseteq \{w'' \mid p_2(w'') \in \{\text{True}, \text{False}\}\}.$
 $\text{CLOS}_{w,EP_w,\text{RES}_c}(\{w'' \mid p_1(w'') = \text{True}\}) \subseteq \{w'' \mid p_2(w'') = \text{True}\}.$

- (120) $CLOS_{w,X,RES_c}(Y)$ is defined iff there is at least one $w' \in (Y \cap X)$ such that:
 $\{w'' \in (Y \cap X) \mid (w, (w'', w')) \in RES_c\} = \emptyset$.
 If defined, $CLOS_{w,X,RES_c}(Y) =$
 $\{w' \in (Y \cap X) \mid \{w'' \in (Y \cap X) \mid (w, (w'', w')) \in RES_c\} = \emptyset\}$.

Accordingly, an indicative conditional *if* p_1 p_2 presupposes, relative to (w, c) , that p_1 has a truth value in w and that all p_1 -worlds in EP_{w_c} that are c -closest to w are worlds where p_2 has a truth value; it asserts that all p_1 -worlds in EP_{w_c} that are c -closest to w are p_2 -worlds.

Let us start with (118a), which has the LF in (121).

- (121) ***if*^{K/P*} $\wedge$ [*Mia has money*] $\wedge$ [*Mia's staff take care of her money*]**

As shown in (122), (119) predicts the intuitive presuppositions of (118a).¹¹

- (122) a. If (121) is CP-compliant relative to c , then:
 $EP_{w_c} \subseteq \{w \mid HAS-MONEY_w(Mia) \in \{True, False\} \text{ and}$
 $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket \textbf{Mia has money} \rrbracket^{w',c} = True\}) \subseteq$
 $\{w' \mid \llbracket \textbf{Mia's staff take care of her money} \rrbracket^{w',c} \in \{True,$
 $False\}\}$.

And so:

- either (I) there is a $w \in EP_{w_c}$ such that:
 $Mia \text{ has money} \sim_w Mia \text{ has a staff}$, and so, although
 $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket \textbf{Mia has money} \rrbracket^{w',c} = True\}) \subseteq \{w^* \mid Mia$
 has a staff in $w^*\}$, Mia need not have a staff in w ,
- or (II) for all $w \in EP_{w_c}$:
 $\neg(Mia \text{ has money} \sim_w Mia \text{ has a staff})$, and so, Mia has a staff
 in w because:
 if Mia has no staff in w , $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket \textbf{Mia has}$
 $\textbf{money} \rrbracket^{w',c} = True\})$ includes some world where Mia has
 no staff (contradicting ' $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket \textbf{Mia has}$
 $\textbf{money} \rrbracket^{w',c} = True\}) \subseteq \{w' \mid \llbracket \textbf{Mia's staff take care of her}$
 $\textbf{money} \rrbracket^{w',c} \in \{True, False\}\}$);
 on the other hand, if Mia has a staff in w , $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid$
 $\llbracket \textbf{Mia has money} \rrbracket^{w',c} = True\}) \subseteq \{w^* \mid Mia \text{ has a staff in } w^*$
 and $\neg(Mia \text{ has money} \sim_{w^*} Mia \text{ has a staff})\}$.

¹¹ Notice that we no longer ignore any s-presuppositions of any predicates. For example, we now acknowledge that *has money* is presuppositional (though we do not specify its s-presuppositions).

- b. If defined, $\llbracket (121) \rrbracket^{w,c} = \text{True}$ iff all the worlds in $\text{CLOS}_{w, \text{EP}_{w_c}, \text{RES}_c}(\{w' \mid \text{Mia has money in } w'\})$ are worlds where Mia's staff take care of Mia's money.

It stands to reason that by default (i.e., unless explicitly established otherwise), the participants of c assume that the laws allow Mia to have money without having a staff (see Pérez Carballo 2009).¹² The result, as shown in (122a-II), is that *Mia has a staff* is true in the worlds of EP_{w_c} .¹³ Those same participants will suppress that default assumption upon hearing *If Mia has money, she has a staff*, because when it is established in the discourse that the laws allow Mia to have money and no staff (in which case Mia is guaranteed to have a staff in all the worlds of EP_{w_c}), $[\text{if}^{K/P*} \wedge [\text{Mia has money}] \wedge [\text{Mia has a staff}]]$ violates CP, due to the un informativity of $[\text{Mia has a staff}]$ (cf. Pérez Carballo [2009]). The LF-based version of CP is adjusted accordingly.

- (123) LF δ is CP-compliant relative to utterance context c only if:
- $\text{EP}_{w_c} \subseteq \{w \mid \llbracket \delta \rrbracket^{w,c} \text{ is defined}\}$,
 - there is no ε such that δ dominates ε and ε is uninformative in δ relative to c .
- (124) ε is uninformative in δ relative to c iff: ε is dominated by δ , and there is a $\pi \in \{\text{tr-}\varepsilon, \text{fl-}\varepsilon\}$ such that for all $w \in \text{EP}_{w_c}$, $\llbracket \delta \rrbracket^{w,c} = \llbracket \delta_{[\varepsilon/\pi]} \rrbracket^{w,c}$, where:
- either (I) ε is not an occurrence of π , in which case $\delta_{[\varepsilon/\pi]}$ is just like δ except that ε is replaced with an occurrence of π ,
 - or (II) ε is an occurrence of π , in which case $\delta_{[\varepsilon/\pi]} = \delta$, and

¹² This, together with the fact that *Mia's staff* and *Mia has money* are sub-LFs of (121), ensures that both $[\lambda w'. \text{Mia has money} \sim_{w'} \text{Mia has a staff}]$ and *Mia has a staff* are salient propositions in c . By assumption, non-salient propositions may be neutralized by RES_c (see Warmbrod (1982) and Stalnaker (1984), where the Limit Assumption is defended, contra Lewis (1973)). And so, for any $q \neq p_1$ that is not salient in c , RES_c may be such that for all w in EP_{w_c} there are worlds w^* and w^{**} such that $q(w) = q(w^*) = \text{True}$ and $q(w^{**}) = \text{False}$, yet both w^* and w^{**} are in $\text{CLOS}_{w, \text{EP}_{w_c}, \text{RES}_c}(\{w'' \mid p_1(w'') = \text{True}\})$. Salience shields from such neutralization.

¹³ Filtering in counterfactual conditionals works in the same way. For example, *If Mia had money, her staff would take care of her money* presupposes (at least) what (118a) presupposes. In the spirit of the Stalnaker/Heim analysis, this may be explained by the assumption that $\text{if}^{K/P*}$ -Counterfactual is just like $\text{if}^{K/P*}$, except for the value of the X-parameter of CLOS (see (120)), which is set to be a proper superset of EP_{w_c} such that: (a) p_1 has a truth value in all the worlds of X , and (b) the laws that hold in the worlds of EP_{w_c} hold in the worlds of X .

- b. for all (w, c') :
 if $\pi \in \{\mathbf{tr}\text{-}\boldsymbol{\varepsilon}, \mathbf{fl}\text{-}\boldsymbol{\varepsilon}\}$, $\llbracket \pi \rrbracket^{w,c'}$ is defined iff $\llbracket \boldsymbol{\varepsilon} \rrbracket^{w,c'} \in \{\text{True}, \text{False}\}$;
 if defined, $\llbracket \mathbf{tr}\text{-}\boldsymbol{\varepsilon} \rrbracket^{w,c'} = \text{True}$ and $\llbracket \mathbf{fl}\text{-}\boldsymbol{\varepsilon} \rrbracket^{w,c'} = \text{False}$.

We turn to (118b,c). Exploiting the assumption that the X parameter of CLOS in (119) is set to be EP_{w_c} , we assume the lexical meanings for **believe** and **know** in (125) and (126).¹⁴ The result is that (118b) violates CP whenever *If Mia has money, she has a staff* violates it, and that (118c) presupposes that Fred believes what (118a) presupposes.

- (125) $\llbracket \mathbf{believe} \rrbracket^{w,c}(p)(z)$ is defined only if for all c' such that EP_{w_c} is the set of worlds compatible with what z believes in w , and all w' in EP_{w_c} , $p(w') \in \{\text{True}, \text{False}\}$.
 If defined, $\llbracket \mathbf{believe} \rrbracket^{w,c}(p)(z) = \text{True}$ iff for all c' such that EP_{w_c} is the set of worlds compatible with what z believes in w , and all w' in EP_{w_c} , $p(w') = \text{True}$.
- (126) $\llbracket \mathbf{know} \rrbracket^{w,c}(p)(z)$ is defined only if $p(w) = \text{True}$ and $\llbracket \mathbf{believe} \rrbracket^{w,c}(p)(z)$ is defined.
 If defined, $\llbracket \mathbf{know} \rrbracket^{w,c}(p)(z) = \llbracket \mathbf{believe} \rrbracket^{w,c}(p)(z)$.

We turn to **and**^{K/P*} and **or**^{K/P*}, which have CLOS-based presuppositions, as in (127)–(128).¹⁵

- (127) When P_1 is a proposition, $\llbracket \mathbf{and}^{K/P*} \rrbracket^{w,c}(P_1)$ is defined iff $\llbracket \mathbf{if}^{K/P*} \rrbracket^{w,c}(P_1)$ is defined.
 If defined, $\llbracket \mathbf{and}^{K/P*} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2: P_2(w) \in \{\text{True}, \text{False}\} \text{ or } \llbracket \mathbf{if}^{K/P*} \rrbracket^{w,c}(P_1)(P_2) \text{ is defined.}]$
 $P_1(w) = P_2(w) = \text{True}]$.
- When P_1 is a property of individuals, $\llbracket \mathbf{and}^{K/P*} \rrbracket^{w,c}(P_1) =$
 $[\lambda P_2. [\lambda x: \llbracket \mathbf{and}^{K/P*} \rrbracket^{w,c}(P_1^x)(P_2^x) \in \{\text{True}, \text{False}\}. \llbracket \mathbf{and}^{K/P*} \rrbracket^{w,c}(P_1^x)(P_2^x)]]$.

¹⁴ The idea that the X parameter of CLOS in (119) is set as EP_{w_c} (rather than EP_w) comes from Domain Semantics (advocated in works such as Veltman [1996], Haegeman [2006] and Yalcin [2007]).

¹⁵ One might object to (127)–(128) on the grounds that **and** and **or** are intuitively non-intensional. The issue is orthogonal to our discussion, and we ignore it in the rest of the paper. It is still worth noting that some connectives that are traditionally considered non-intensional have been argued to have an intensional meaning, even at the assertion tier (see Kratzer [1989], Zimmermann [2000], Berto [2015], Abrusán [2023]). (127)–(128) could be fully intensionalized in that spirit. Such intensionalization would also eliminate the following discrepancy. When p_1 is a contradiction, *if* p_1 p_2 comes out a presupposition failure by (119), but p_1 **and** p_2 and *not*- p_1 **or** p_2 merely violate CP by (127)–(128).

- (128) When P_1 is a proposition, $\llbracket \text{or}^{K/P*} \rrbracket^{w,c}(P_1) =$
 $\llbracket \lambda P_2: (\llbracket \text{and}^{K/P*} \rrbracket^{w,c}(P_1) \text{ is defined or } \llbracket \text{and}^{K/P*} \rrbracket^{w,c}(P_2) \text{ is defined}) \text{ and}$
 $\llbracket \text{and}^{K/P*} \rrbracket^{w,c}([\lambda w': P_1(w') \text{ is defined. } P_1(w') = \text{False}]) (P_2) \text{ is defined and}$
 $\llbracket \text{and}^{K/P*} \rrbracket^{w,c}([\lambda w': P_2(w') \text{ is defined. } P_2(w') = \text{False}]) (P_1) \text{ is defined.}$
 $P_1(w) = \text{True or } P_2(w) = \text{True}]$.
 When P_1 is a property of individuals, $\llbracket \text{or}^{K/P*} \rrbracket^{w,c}(P_1) =$
 $\llbracket \lambda P_2. [\lambda x: \llbracket \text{or}^{K/P*} \rrbracket^{w,c}(P_1^x)(P_2^x) \in \{\text{True}, \text{False}\}. \llbracket \text{or}^{K/P*} \rrbracket^{w,c}(P_1^x)(P_2^x)] \rrbracket$.
- (129) $Z^x = [\lambda w': Z(w')(x) \in \{\text{True}, \text{False}\}. Z(w')(x)]$

Thus, the context-dependency of filtering in conjunctions and disjunctions, including conjunctions and disjunctions embedded under *believe*, is also explained. Indeed, on their 'de dicto' readings, (130a) presupposes that Fred believes what *Mia has money and her staff take care of her money* presupposes, and (130b) presupposes that Fred believes what *Mia has no money or her staff take care of her money* presupposes.

- (130) a. Fred believes that Mia has money and her staff take care of her money.
 b. Fred believes that (either) Mia has no money or her staff take care of her money.

In addition, with (123)–(124), redundancy contrasts are also accounted for (cf. Section 2). This includes the fact that construed *de dicto*, (131a) and (131b) are inherently odd, but (131c) is not inherently odd (and is equivalent to (130b)).

- (131) a. Fred believes that if Mia's staff take care of her money, she has money.
 b. Fred believes that Mia's staff take care of her money and she has money.
 c. Fred believes that (either) Mia's staff take care of her money or she has no money.

Now, suppose both global and non-global accommodation are allowed in principle, even in K/P^* . Accordingly, in addition to (121), (118a) has the LFs in (132); (132a) is the strongest.

- (132) a. $\text{and}^{K/P*} \wedge [\text{Mia has a staff and } \text{can in principle have money}] \wedge [\text{if}^{K/P*}$
 $\wedge [\text{Mia has money}] \wedge [\text{Mia's staff take care of her money}]]$
 b. $\text{if}^{K/P*} \wedge [\text{Mia has money}] \wedge [\text{and}^{K/P*} \wedge [\text{Mia has a staff}] \wedge [\text{Mia's staff take}$
 $\text{care of her money}]]$

This raises the question of whether K/P^* has an advantage over K/P regarding (118a) at all, as K/P generates for (118a) a parallel triple of LFs, namely, ((106), (108a), (108b)). The answer should be obvious by now: even if K/P^* has, in fact, no advantage over K/P regarding (118a), it can still have an advantage over K/P regarding Rooth-Partee conditionals. Specifically, since – as we saw in 4.2 – K/P fails to account for the $\forall$ -over-

if reading of the Rooth-Partee conditional (133), the pertinent question is: can K/P* account for both readings of (133)?

(133) If Mia is penniless or proud of her money, then Sue is (too).

(134) a. Intuitive if-over- $\exists$ presupposition of (133) (see (88)):

Mia and Sue can have money, in principle.

b. Intuitive if-over- $\exists$ assertion of (133):

If Mia is penniless or proud of Mia's money, Sue is penniless or proud of Sue's money.

(135) a. Intuitive $\forall$ -over-if presupposition of (133) (see (89); cf. (91)):

(i) Mia can have money, in principle, and

(ii) Sue can have money in principle, provided that the (pragmatically) presupposed laws allow Mia to be penniless while Sue cannot have money in principle,¹⁶

(iii) if Mia has money, Sue has money, provided that the (pragmatically) presupposed laws allow Mia to be proud of her money while Sue has no money.

b. Intuitive $\forall$ -over-if assertion of (133):

If Mia is penniless, Sue is penniless, and if Mia has money and is proud of Mia's money, Sue is proud of Sue's money.

Within K/P*, the accommodation-free (136) is an LF of (133). (137) provides the possible values of *prop_E*, and (138) provides the meanings of *penniless* and *proud of her money*.

(136) $if^{K/P*} \wedge [Mia [or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money]]] \wedge [Sue \vee prop_E]$

(137) a. *prop_E* denotes $f1prop_E = \llbracket [or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money]] \rrbracket^{w,c}$

b. *prop_E* denotes $f2prop_E$, which is such that for all w:

if $\llbracket penniless \rrbracket^{w,c}(Mia) = \text{True}$, $f2prop_E(w) = \llbracket penniless \rrbracket^{w,c}$,

if $\llbracket proud\ of\ her\ money \rrbracket^{w,c}(Mia) = \text{True}$, $f2prop_E(w) = \llbracket proud\ of\ her\ money \rrbracket^{w,c}$.

(138) a. $\llbracket penniless \rrbracket^{w,c} =$

$[\lambda x: HAS-MONEY_w(x) \in \{\text{True}, \text{False}\}. HAS-MONEY_w(x) = \text{False}]$

b. $\llbracket proud\ of\ her\ money \rrbracket^{w,c} =$

$[\lambda x: HAS-MONEY_w(x) = \text{True} \text{ and } PROUD-OF_w(x, x's\ money) \in \{\text{True}, \text{False}\}. PROUD-OF_w(x, x's\ money) = \text{True}]$

¹⁶ It is hard to imagine circumstances where the laws would disallow this. This is why, in practice, the $\forall$ -over-if reading of (133) intuitively presupposes that Sue can – in principle – have money.

Pairing (136) with $f1prop_E$ derives the *if-over- $\exists$* meaning in (139). Pairing (136) with $f2prop_E$ derives the meaning in (140), which does not account for the intuitive $\forall$ -over-*if* reading of (133) described in (135). Contra (135)+CP, (140) is compatible with Scenario I in (141), where *Mia is penniless* and *Mia is proud of her money* may have different likelihoods.¹⁷

(139) Pairing (136) with $f1prop_E$ (if-over- $\exists$)

- a. If (136) is CP-compliant, then $EP_{w_c} \subseteq$
 $\{w \mid \llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w,c}(Mia) \in \{\text{True}, \text{False}\},$
and
 $(\llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w,c}(Sue) \in \{\text{True}, \text{False}\} \text{ or}$
for all w^* in $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w',c}(Mia) = \text{True}\})$: $f1prop_E(w^*)(Sue) \in \{\text{True}, \text{False}\})$.
- b. If defined, $\llbracket (136) \rrbracket^{w,c} = \text{True}$ iff for all w in $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \text{Mia is penniless or proud of Mia's money in } w'\})$, Sue is penniless or proud of Sue's money in w .

(140) Pairing (136) with $f2prop_E$ (not $\forall$ -over-*if*)

- a. If (136) is CP-compliant, then $EP_{w_c} \subseteq$
 $\{w \mid \llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w,c}(Mia) \in \{\text{True}, \text{False}\},$
and for all (w^*, Y) such that w^* is in $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w',c}(Mia) = \text{True}\})$ and Y is in $\{\llbracket penniless \rrbracket^{w^*,c}, \llbracket proud\ of\ her\ money \rrbracket^{w^*,c}\}$:
 $Y(Mia) = \text{False}$ or $f2prop_E(w^*)(Sue) \in \{\text{True}, \text{False}\}$.
- b. If defined, $\llbracket (136) \rrbracket^{w,c} = \text{True}$ iff for all w in $CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \text{Mia is penniless or proud of Mia's money in } w'\})$: if Mia is penniless in w , Sue is penniless in w , and if Mia is proud of Mia's money in w , Sue is proud of Sue's money in w .

(141) Scenario I

- a. $\{w \in EP_{w_c} \mid \text{Mia has money in } w\} \subseteq$
 $\{w \mid \neg(\text{Mia is proud of her money} \rightsquigarrow_w \text{Sue has money})\}$
- b. $EP_{w_c} \cap \{w \mid \text{it is more realistic in } w \text{ for Mia to be penniless than to have money, and Mia has money in } w, \text{ and Mia is not proud of her money in } w, \text{ and Sue has no money in } w, \text{ and } \{w' \mid \llbracket penniless \rrbracket^{w',c}(Mia) = \text{True}\} \supseteq CLOS_{w,EP_{w_c},RES_c}(\{w' \mid \llbracket or^{K/P*} \wedge penniless \wedge [proud\ of\ her\ money] \rrbracket^{w',c}(Mia) = \text{True}\})\} \neq \emptyset$

¹⁷ An additional mechanism—as in Alonso-Ovalle (2009), Santorio (2018) or Bar-Lev and Fox (2020) — may be needed to guarantee SDA (Simplification of Disjunctive Antecedents) in (139).

- c. $EP_{w_c} \cap \{w' \mid \llbracket \text{penniless} \rrbracket^{w',c}(\text{Mia}) = \text{True} \} \not\subseteq \text{CLOS}_{w,EP_{w_c},\text{RES}_c}(\{w' \mid \llbracket \text{or}^{K/P^*} \text{penniless} \wedge [\text{proud of her money}] \rrbracket^{w',c}(\text{Mia}) = \text{True} \}) \neq \emptyset$

The problem arises due to the fact that (140a) references the set of *Mia is penniless or proud of her money*-worlds closest to w , instead of referencing – separately – the set of *Mia is penniless*-worlds closest to w and the set of *Mia is proud of her money*-worlds closest to w . Therefore, we replace K/P^* with K/P^{**} , which allows *if* to quantify over the properties that are the arguments of *or* (as in Rooth and Partee (1982)). In K/P^{**} (i) the type-flexible $\text{or}^{K/P^{**}}$ in (142a) replaces the type-flexible or^{K/P^*} (cf. Abenina-Adar and Sharvit [2021]), and (ii) the type-flexible $\text{if}^{K/P^{**}}$ in (142b) replaces the type-rigid if^{K/P^*} .

- (142) a. When Q_1 is a proposition, $\llbracket \text{or}^{K/P^{**}} \rrbracket^{w,c}(Q_1) = \llbracket \text{or}^{K/P^*} \rrbracket^{w,c}(Q_1)$.
When Q_1 is a property of individuals, $\llbracket \text{or}^{K/P^{**}} \rrbracket^{w,c}(Q_1) =$
 $\llbracket \lambda Q_2. \llbracket \text{or}^{K/P^{**}} \rrbracket^{w,c}(Q_1^x)(Q_2^x) \in \{\text{True}, \text{False}\}. \llbracket \lambda Z. (Z = Q_1 \text{ or } Z = Q_2).$
 $Z(w)(x)) \rrbracket \rrbracket$
- b. When O_1 is a proposition, $\llbracket \text{if}^{K/P^{**}} \rrbracket^{w,c}(O_1) = \llbracket \text{if}^{K/P^*} \rrbracket^{w,c}(O_1)$.
When O_1 is a function from worlds to functions from properties of individuals to truth values, $\llbracket \text{if}^{K/P^{**}} \rrbracket^{w,c}(O_1)$ is defined iff $\{Z \mid O_1(w)(Z) \in \{\text{True}, \text{False}\} \neq \emptyset\}$, and if defined, $\llbracket \text{if}^{K/P^{**}} \rrbracket^{w,c}(O_1) =$
 $\llbracket \lambda O_2. \{Z \mid O_1(w)(Z) \in \{\text{True}, \text{False}\} \subseteq$
 $\{Z \mid \text{CLOS}_{w,EP_{w_c},\text{RES}_c}(\{w' \mid O_1(w')(Z) = \text{True}\}) \subseteq$
 $\{w' \mid O_2(w')(Z) \in \{\text{True}, \text{False}\}\} \rrbracket$.
 $\{Z \mid O_1(w)(Z) \in \{\text{True}, \text{False}\} \subseteq$
 $\{Z \mid \text{CLOS}_{w,EP_{w_c},\text{RES}_c}(\{w' \mid O_1(w')(Z) = \text{True}\}) \subseteq \{w' \mid O_2(w')(Z) = \text{True}\}\} \rrbracket$.

The LFs in (143) are generated for (133) within K/P^{**} . Their interpretations rely on the additional rules in (144), where for all contexts c , g_c is the variable assignment of c .

- (143) a. $\text{if}^{K/P^{**}} \wedge [\exists [\text{Mia} [\text{or}^{K/P^{**}} \wedge \text{penniless} \wedge [\text{proud of her money}]]]] \wedge [\exists [\text{Sue} \vee \text{prop}_E]]$
b. $\text{if}^{K/P^{**}} \wedge [\text{Mia} [\text{or}^{K/P^{**}} \wedge \text{penniless} \wedge [\text{proud of her money}]]] \wedge [\text{i} [\text{Sue} \vee \text{prop}_i]]$
- (144) a. $\llbracket \exists \rrbracket^{w,c} = [\lambda P: \{X \mid P(X) \in \{\text{True}, \text{False}\} \neq \emptyset. \{X \mid P(X) = \text{True}\} \neq \emptyset]$
b. $\llbracket \text{i } \delta \rrbracket^{w,c} = [\lambda U: \llbracket \delta \rrbracket^{w,c[i \rightarrow U]} \text{ is defined. } \llbracket \delta \rrbracket^{w,c[i \rightarrow U]}],$
where $g_{c[i \rightarrow U]}$ maps i to U , and $c[i \rightarrow U]$ is otherwise just like c
(see Heim and Kratzer (1998))
c. If y is a pronoun, $\llbracket y_i \rrbracket^{w,c}$ is defined only if $g_c(\text{i})$ is defined.
If defined, $\llbracket y_i \rrbracket^{w,c} = g_c(\text{i})$.

Relative to (w, c) , $[or^{K/P^{**}} \wedge penniless \wedge [proud of her money]]$ denotes a function f from individuals to functions from properties of individuals to truth values such that for any x , $Dom(f(x)) \subseteq \{[\wedge penniless]^{w,c}, [\wedge proud of her money]^{w,c}\}$. In (143a), the ellipsis site is filled by $\vee prop_E$, where $prop_E$ is anaphoric to $[\wedge [or^{K/P^{**}} \wedge penniless \wedge [proud of her money]]]^{w,c}$. In (143b), the ellipsis site is filled by the i -bound $\vee prop_i$, and $[i [Sue is \vee prop_i]]^{w,c} = [\lambda P: P(w)(Sue) \in \{True, False\}. P(w)(Sue)]$. As shown in (145)–(146), when $\alpha = \wedge penniless$ and $\beta = \wedge proud of her money$, (143a) – where $if^{K/P^{**}}$ “takes” propositions – yields the *if-over- $\exists$* meaning of (136), described in (134), and (143b) – where $if^{K/P^{**}}$ “takes” functions from worlds to functions from properties of individuals to truth values – yields its *$\forall$ -over-if* meaning, described in (135).

- (145) a. If (143a) is CP-compliant relative to c , then:
 $EP_{w_c} \subseteq \{w | \{Z | [or^{K/P^{**}} \alpha \beta]^{w,c}(Mia)(Z) \in \{True, False\}\} \neq \emptyset \text{ and } CLOS_{w, EP_{w_c}, RES_c}(\{w' | \{Z | [or^{K/P^{**}} \alpha \beta]^{w',c}(Mia)(Z) = True\} \neq \emptyset\}) \subseteq \{w' | \{Z | [or^{K/P^{**}} \alpha \beta]^{w',c}(Sue)(Z) \in \{True, False\}\} \neq \emptyset\}\}$
 b. If defined, $[(143a)]^{w,c} = True$ iff
 $CLOS_{w, EP_{w_c}, RES_c}(\{w | [\alpha]^{w,c}(w)(Mia) = True \text{ or } [\beta]^{w,c}(w)(Mia) = True\}) \subseteq \{w | [\alpha]^{w,c}(w)(Sue) = True \text{ or } [\beta]^{w,c}(w)(Sue) = True\}$.
- (146) a. If (143b) is CP-compliant relative to c , then:
 $EP_{w_c} \subseteq \{w | \{Z | [or^{K/P^{**}} \alpha \beta]^{w,c}(Mia)(Z) \in \{True, False\}\} \neq \emptyset \text{ and for all } Z \text{ such that } [or^{K/P^{**}} \alpha \beta]^{w,c}(Mia)(Z) \in \{True, False\}, CLOS_{w, EP_{w_c}, RES_c}(\{w' | Z(w')(Mia) = True\}) \subseteq \{w' | Z(w')(Sue) \in \{True, False\}\}\}$
 And so, $M_c \neq \emptyset$, where $M_c = \{w' \in EP_{w_c} | Mia \text{ has money in } w'\}$, and (cf. (122a)):
 either (I) $M_c \cap \{w' | Mia \text{ is proud of her money} \rightsquigarrow_{w'} Sue \text{ has money}\} \neq \emptyset$, and so, some w in M_c might be such that although
 $CLOS_{w, EP_{w_c}, RES_c}(\{w' | [Mia \text{ is proud of her money}]^{w',c} = True\}) \subseteq \{w^* | Sue \text{ has money in } w^*\}$, Sue has no money in w ;
 or (II) $M_c \cap \{w' | Mia \text{ is proud of her money} \rightsquigarrow_{w'} Sue \text{ has money}\} = \emptyset$, and so, Sue has money in all the worlds of M_c , because for all w in M_c :
 if Sue is penniless in w , $CLOS_{w, EP_{w_c}, RES_c}(\{w' | [proud of her money]^{w',c}(Mia) = True\})$ includes worlds where Sue is penniless (contradicting ' $CLOS_{w, EP_{w_c}, RES_c}(\{w' | [proud of her money]^{w',c}(Mia) = True\}) \subseteq \{w'' | [proud of her money]^{w'',c}(Sue) \in \{True, False\}\}$);
 on the other hand, if Sue has money in w , $CLOS_{w, EP_{w_c}, RES_c}(\{w' | [proud of her money]^{w',c}(Mia) = True\}) \subseteq \{w^* | Sue \text{ has money in } w^* \text{ and } \neg(Mia \text{ is proud of her money} \rightsquigarrow_{w^*} Sue \text{ has money})\}$.

- b. If defined, $\llbracket (143b) \rrbracket^{w,c} = \text{True}$ iff for all Z such that $\llbracket \text{or}^{K/P^{**}} a \beta \rrbracket^{w,c}(Mia)(Z) \in \{\text{True}, \text{False}\}$: $\text{CLOS}_{w,EP_{w,c},\text{Res}_c}(\{w \mid Z(w)(Mia) = \text{True}\}) \subseteq \{w \mid Z(w)(Sue) = \text{True}\}$.

Since $\text{or}^{K/P^{**}}$ is symmetric, (143a,b) also account for both readings of *If Mia is proud of her money or penniless, then Sue is*.

6 Two additional consequences

We have argued that *or* is inherently symmetric. Indeed, the alternative to K/P^{**} in (147) fails to replicate the predictions of K/P^{**} : (148a) and (148b) are intuitively equivalent, but since both if^{ALT} and or^{ALT} are asymmetric, (149a) and (149b) have different meanings.

- (147) a. When P_1 is a proposition, $\llbracket \text{or}^{\text{ALT}} \rrbracket^{w,c}(P_1)$ is defined iff $P_1(w)$ is defined.
 If defined, $\llbracket \text{or}^{\text{ALT}} \rrbracket^{w,c}(P_1) = [\lambda P_2: P_2(w) \text{ is defined or } \llbracket \text{if}^{\text{ALT}} \rrbracket^{w,c}([\lambda w': P_1(w') \text{ is defined. } P_1(w') = \text{False}]) (P_2) \text{ is defined. } P_1(w) = \text{True or } P_2(w) = \text{True}]$.
 When P_1 is a property of individuals, $\llbracket \text{or}^{\text{ALT}} \rrbracket^{w,c}(P_1) = [\lambda P_2. [\lambda x: \llbracket \text{or}^{\text{ALT}} \rrbracket^{w,c}(P_1^x)(P_2^x) \text{ is defined. } [\lambda Z: (Z = P_1 \text{ or } Z = P_2). Z(w)(x)]]]$
- b. $\llbracket \text{if}^{\text{ALT}} \rrbracket^{w,c} = \llbracket \text{if}^{K/P^{**}} \rrbracket^{w,c}$ (see (142b))
- (148) a. If Mia is ashamed of her children or proud of her money, then Sue is.
 b. If Mia is proud of her money or ashamed of her children, then Sue is.
- (149) a. $\text{if}^{\text{ALT}} \wedge [\text{Mia } [\text{or}^{\text{ALT}} \chi \eta]] \wedge [\text{i } [\text{Sue } \vee \text{prop}_i]]$
 Presupposes what $[\text{Mia } \vee \chi]$ presupposes,
 but not necessarily what $[\text{Mia } \vee \eta]$ presupposes.
- b. $\text{if}^{\text{ALT}} \wedge [\text{Mia } [\text{or}^{\text{ALT}} \eta \chi]] \wedge [\text{i } [\text{Sue } \wedge \text{prop}_i]]$
 Presupposes what $[\text{Mia } \vee \eta]$ presupposes,
 but not necessarily what $[\text{Mia } \vee \chi]$ presupposes.

As it happens, (149a) and (149b) are Strawson-equivalent; that is to say, their denotations Strawson-entail each other in the sense of von Fintel 1999 (p *Strawson-entails* q iff q is entailed by the conjunction of p with the presuppositions of q). This does not make (147) a viable alternative to K/P^{**} , however, because the mere Strawson-equivalence of (149a) and (149b) cannot explain why (148a) and (148b) have the same intuitive $\vee$ -over-*if*-reading. To appreciate the problem, compare (148a,b) to (150a,b). The LFs of (150a) and (150b) are Strawson-equivalent, but (150a) and (150b)

are intuitively non-equivalent in the classical sense, while (148a) and (148b) are intuitively equivalent in the classical sense.

- (150) a. The winner cried. Presupposes: there is exactly one winner.
 b. The French winner cried. Presupposes: there is exactly one French winner.

The alternative to K/P^{**} in (147) happens to be similarity-based, but since a satisfactory account of (148a) and (148b) requires that they have classically equivalent LFs, it seems fair to say that any theory TH – similarity-based or not – accounts for (148a) and (148b) only if or^{TH} is symmetric. The non-similarity-based alternative in Winter (2025) is a case in point.

A note on type-flexibility. In K/P^{**} , (151a) has an LF where $if^{K/P^{**}}$ quantifies over the property-arguments of $or^{K/P^{**}}$. One might be inclined to think, with Stone (1992), that (151b) – which has clausal disjuncts – favors K/P^* , where if^{K/P^*} is strictly clause-taking.

- (151) a. If Mia is penniless or proud of her money, then Sue is (too).
 b. If Mia is penniless or Ann is proud of her money, then Sue is (too).

However, referencing the set of closest *Mia is penniless or Ann is proud of her money*-worlds would be as problematic for (151b) – which has the meaning in (152) – as referencing the set of closest *Mia is penniless or proud of her money*-worlds is for (151a) (see the discussion of (140) in Section 5).

- (152) a. Intuitive presupposition of (151b), simplified:
 (i) Ann has money, provided that the (pragmatically) presupposed laws allow Mia to have money while Ann has no money, and
 (ii) if Ann has money, Sue has money, provided that the (pragmatically) presupposed laws allow Ann to be proud of her money while Sue has no money.
 b. Intuitive assertion of (151b):
 If Mia is penniless, Sue is penniless, and if Ann has money and is proud of her money, Sue is proud of her money.

Therefore, to account for Rooth-Partee conditionals with clausal disjuncts, within K/P^{**} , we assume that the type-flexibility enjoyed by $if^{K/P^{**}}$ and $or^{K/P^{**}}$ subsumes (142) but goes beyond it. Accordingly, $if^{K/P^{**}}$ may quantify over the arguments of $or^{K/P^{**}}$ when those arguments denote structured propositions. Thus, for example, (151b) has the LF in (153), where the arguments of $or^{K/P^{**}}$ denote individual-property pairs. The first argument of $if^{K/P^{**}}$ in (153) denotes a function from worlds to functions from individual-property pairs to truth values; this follows from (142a)+(154). By (142b)+(155), the meaning of (153) is as in (156).

- (153) $if^{K/P^{**}} \wedge [or^{K/P^{**}} [Mia \clubsuit a] [Ann \clubsuit \beta]] \wedge [i [Sue \vee prop_i]]$
- (154) a. $\llbracket or^{K/P^{**}} [Mia \clubsuit a] [Ann \clubsuit \beta] \rrbracket^{w,c} =$
 $[\lambda Y: \llbracket or^{K/P^{**}} \rrbracket^{w,c} ([\llbracket a \rrbracket^{w,c}]^{Mia}) ([\llbracket \beta \rrbracket^{w,c}]^{Ann}) \in \{\text{True}, \text{False}\} \text{ and}$
 $(Y = (Mia, [\llbracket a \rrbracket^{w,c}]) \text{ or } Y = (Ann, [\llbracket \beta \rrbracket^{w,c}])). [2nd(Y)](w)(1st(Y))]$
 b. $Z^x = [\lambda w': Z(w')(x) \in \{\text{True}, \text{False}\}. Z(w')(x)]$
 c. $1st(b, B) = b$, and $2nd(b, B) = B$
 d. $\llbracket p \clubsuit \sigma \rrbracket^{w,c} = ([\llbracket p \rrbracket^{w,c}], [\llbracket \sigma \rrbracket^{w,c}])$
 e. When Q_1 is an individual-property pair, $\llbracket or^{K/P^{**}} \rrbracket^{w,c}(Q_1) =$
 $[\lambda Q_2: \llbracket or^{K/P^{**}} \rrbracket^{w,c} ([2nd(Q_1)]^{1st(Q_1)}) ([2nd(Q_2)]^{1st(Q_2)}) \in \{\text{True}, \text{False}\}.]$
 $[\lambda Y: (Y = Q_1 \text{ or } Y = Q_2). [2nd(Y)](w)(1st(Y))]]$
- (155) When O_1 is a function from worlds to functions from individual-property pairs to truth values, and O_2 is a function from worlds to functions from properties to truth values, $\llbracket if^{K/P^{**}} \rrbracket^{w,c} =$
 $[\lambda O_1: \{Y \mid O_1(w)(Y) \in \{\text{True}, \text{False}\}\} \neq \emptyset.$
 $[\lambda O_2: \{Y \mid O_1(w)(Y) \in \{\text{True}, \text{False}\} \subseteq$
 $\{Y \mid CLOS_{w, EP_{w,c}, RES_c}(\{w' \mid [2nd(Y)](w')(1st(Y)) = \text{True}\}) \subseteq$
 $\{w' \mid O_2(w')(2nd(Y)) \in \{\text{True}, \text{False}\}\}.$
 $\{Y \mid O_1(w)(Y) \in \{\text{True}, \text{False}\} \subseteq$
 $\{Y \mid CLOS_{w, EP_{w,c}, RES_c}(\{w' \mid [2nd(Y)](w')(1st(Y)) = \text{True}\}) \subseteq$
 $\{w' \mid O_2(w')(2nd(Y)) = \text{True}\}\}]]]$
- (156) If (153) is CP-compliant relative to c , then:
 $EP_{w,c} \subseteq \{w \mid \{Y \mid \llbracket or^{K/P^{**}} [Mia \clubsuit a] [Ann \clubsuit \beta] \rrbracket^{w,c}(Y) \in \{\text{True}, \text{False}\}\} \neq \emptyset,$
 and for all Y such that $\llbracket or^{K/P^{**}} [Mia \clubsuit a] [Ann \clubsuit \beta] \rrbracket^{w,c}(Y) \in \{\text{True}, \text{False}\}:$
 $CLOS_{w, EP_{w,c}, RES_c}(\{w' \mid [2nd(Y)](w')(1st(Y)) = \text{True}\}) \subseteq$
 $\{w' \mid [2nd(Y)](w')(Sue) \in \{\text{True}, \text{False}\}\}.$
 If defined, $\llbracket (153) \rrbracket^{w,c} = \text{True}$ iff
 for all Y such that $\llbracket or^{K/P^{**}} [Mia \clubsuit a] [Ann \clubsuit \beta] \rrbracket^{w,c}(Y) \in \{\text{True}, \text{False}\}:$
 $CLOS_{w, EP_{w,c}, RES_c}(\{w \mid [2nd(Y)](w)(1st(Y)) = \text{True}\}) \subseteq$
 $\{w \mid [2nd(Y)](w)(Sue) = \text{True}\}.$

This version of K/P^{**} offers (157a) and (157b), where the disjuncts are clausal (and for any δ , $\mathfrak{s}$ is an unpronounced copy of δ), as alternative *if-over- $\exists$* and *$\forall$ -over-if* LFs of (151a).

- (157) a. $if^{K/P^{**}} \wedge [or^{K/P^{**}} [Mia \vee a] [\mathfrak{s}ia \vee \beta]] \wedge [or^{K/P^{**}} [Sue \vee prop_E] [\mathfrak{s}ue \vee prop_E]]$
 b. $if^{K/P^{**}} \wedge [or^{K/P^{**}} [Mia \clubsuit a] [\mathfrak{s}ia \clubsuit \beta]] \wedge [i [Sue \vee prop_i]]$

For empirical arguments supporting the thesis that all disjuncts are underlyingly clausal, see Hirsch (2017) and Schein (2017).¹⁸

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References

- Abenina-Adar, Maayan & Yael Sharvit. 2021. On the presuppositional strength of interrogative clauses. *Natural Language Semantics* 29. 47–90.
- Abrusán, Márta. 2023. Projection, attention and the meaning of negation. Handout of SLIME 2. (held at UCLA).
- Alonso-Ovalle, Luis. 2009. Counterfactuals, correlatives, and disjunction. *Linguistics and Philosophy* 32(2). 207–244.
- Bar-Lev, Moshe E and Danny, Fox. 2020. Free choice, simplification, and innocent inclusion. *Natural Language Semantics* . 28(3). 175–223.
- Beaver, David & Emiel Krahmer. 2001. Presupposition and partiality: Back to the future. *Journal of Logic, Language and Information* 10(2). 147–182.
- Berto, Francesco. 2015. A modality called ‘negation’. *Mind* 124. 761–793.
- Chemla, Emmanuel & Philippe Schlenker. 2012. Incremental versus symmetric accounts of presupposition projection: An experimental approach. *Natural Language Semantics* 20. 177–226.
- Gazdar, Gerald. 1979. *Pragmatics, implicature, presupposition, and logical form*. New York: Academic Press.
- Geurts, Bart. 1996. Non-global satisfaction guaranteed: A presupposition theory and its problems. *Linguistics and Philosophy* 19. 259–294.
- Geurts, Bart. 1999. *Presuppositions and pronouns*. New York: Elsevier.
- Grice, Paul. 1989. *Studies in the way of words*. Cambridge, MA: Harvard University Press.
- Hacquard, Valentine. 2006. *Aspects of modality*. Cambridge, MA: Massachusetts Institute of Technology PhD dissertation.
- Heim, Irene. 1982. *The semantics of definite and indefinite noun phrases*. Amherst, MA: University of Massachusetts at Amherst PhD dissertation.
- Heim, Irene. 1983. On the projection problem for presuppositions. *WCCFL* 2. 114–125.

18 Some of our consultants find (i-a) below acceptable and its ellipsis-free counterpart (i-b) odd. The only reading of (i-a) that is true, given the set-up, asserts ‘either *If Mia is married, Sue is married* is true or *If Mia is dating, Sue is dating* is true’. This suggests that *if* is not only type-flexible, but also quantificationally flexible.

- (i) Both Sam and I knew that if Mia is married but not dating, Sue is neither married nor dating. What I knew, but Sam didn’t, is that if Mia is dating, Sue is dating. Not wanting to reveal everything I knew, I said to him:
- a. If Mia is married or dating, then Sue is.
 - b. #If Mia is married or dating, then Sue is married or dating.

- Heim, Irene. 1990. E-type pronouns and donkey anaphora. *Linguistics and Philosophy* 13. 137–177.
- Heim, Irene. 1992. Presupposition projection and the semantics of attitude verbs. *Journal of Semantics* 9. 183–221.
- Heim, Irene & Angelika Kratzer. 1998. *Semantics in generative grammar*. Hoboken, New Jersey: Wiley-Blackwell.
- Hirsch, Aron. 2017. *An inflexible semantics for cross-categorical operators*. Cambridge, MA: Massachusetts Institute of Technology PhD dissertation.
- Hirsch, Aron & Martin Hackl. 2014. Incremental presupposition evaluation in disjunction. In Jyoti Iyer & Leland Kusmer (eds.), *Proceedings of NELS*, vol. 44, 177–190. GLSA Publications.
- Kalomirois, Aleksandros & Florian Schwarz. 2024. Presupposition projection from ‘and’ and ‘or’: Experimental data and theoretical implications. *Journal of Semantics* 41(3–4). 331–372.
- Karttunen, Lauri. 1973. Presuppositions of compound sentences. *Linguistic Inquiry* 4. 169–193.
- Karttunen, Lauri. 1974. Presuppositions and linguistic context. *Theoretical Linguistics* 1. 181–193.
- Karttunen, Lauri & Stanley Peters. 1979. Conventional implicature. In Choon-Kyu Oh & David Dinneen (eds.), *Syntax and semantics*, vol. 11, 1–56. New York: Academic Press.
- Kratzer, Angelika. 1989. An investigation of the lumps of thought. *Linguistics and Philosophy* 12. 607–653.
- Lewis, David. 1973. *Counterfactuals*. Cambridge, MA: Harvard University Press.
- Lewis, David. 1979. Scorekeeping in a language game. *Journal of Philosophical Logic* 8(1). 339–359.
- Mandelkern, Matthew. 2016. A note on the architecture of presupposition. *Semantics and Pragmatics* 9. 1–24.
- Mandelkern, Matthew, Jeremy Zehr, Jacopo Romoli & Florian Schwarz. 2020. We’ve discovered that projection across conjunction is asymmetric (and it is!). *Linguistics and Philosophy* 43. 473–514.
- Partee, Barbara & Mats Rooth. 1982. Generalized conjunction and type-ambiguity. In Rainer Bauerle, Christoph Schwarze & Arnim von Stechow (eds.), *Meaning, use and interpretation of language*, 361–383. Berlin: Walter de Gruyter.
- Pearson, Hazel. 2010. A modification of the “Hey, wait a minute” test. *Snippets* 22. 7–8.
- Pérez Carballo, Alejandro. 2009. Toward a dissolution of the proviso problem. In Paul Égré & Giorgio Magri (eds.), *MIT working papers in linguistics*, vol. 60, 169–184.
- Roberts, Craige & Mandy Simons. 2024. Preconditions and projection: Explaining non-anaphoric presupposition. *Linguistics and Philosophy* 47. 703–748.
- Rooth, Mats & Barbara Partee. 1982. Conjunction, type ambiguity and wide scope *or*. In Daniel P. Flickinger, Marlys Macken & Nancy Wiegand (eds.), *Proceedings of WCCFL*, vol. 1.
- Rothschild, Daniel. 2011/2015. Explaining presupposition projection with dynamic semantics. *Semantics and Pragmatics* 4(3). 1–43.
- Santorio, Paolo. 2018. Alternatives and truthmakers in conditional semantics. *The Journal of Philosophy* 115(10). 513–549.
- Schein, Barry. 2017. *‘And’: Conjunction reduction redux*. Cambridge, MA: Massachusetts Institute of Technology Press.
- Schlenker, Philippe. 2008. Be articulate: A pragmatic theory of presupposition projection. *Theoretical Linguistics* 34. 157–212.
- Schlenker, Philippe. 2009. Local contexts. *Semantics and Pragmatics* 2(3). 1–74.
- Schlenker, Philippe. 2011. Presupposition projection: Two theories of non-global contexts (parts I & II). *Language and Linguistics Compass* 5. 848–879.
- Soames, Scott. 1979. A projection problem for speaker presupposition. *Linguistic Inquiry* 10. 623–666.
- Soames, Scott. 1989. Presupposition. In Dov Gabbay & Franz Guenther (eds.), *Handbook of philosophical logic IV*, 553–616.

- Stalnaker, Robert. 1974. Pragmatic presuppositions. In Milton Munitz & Peter Unger (eds.), *Semantics and philosophy*, 197–213. New York: NYU Press.
- Stalnaker, Robert. 1975. Indicative conditionals. *Philosophia* 5. 269–286.
- Stalnaker, Robert. 1984. *Inquiry*. Cambridge, MA: Massachusetts Institute of Technology Press.
- Stone, Matthew. 1992. *Or* and anaphora. In *Proceedings of SALT*, vol. II, 367–385.
- Veltman, Frank. 1996. Defaults in update semantics. *Journal of Philosophical Logic* 25(3). 221–261.
- von Fintel, Kai. 1999. NPI licensing, Strawson entailment, and context dependency. *Journal of Semantics* 16. 97–148.
- von Fintel, Kai. 2004. Would you believe it? The king of France is back! Presuppositions and truth-value intuitions. In Marga Reimer & Anne Bezuidenhout (eds.), *Descriptions and beyond*, 315–341. Oxford: Oxford University Press.
- Warmbrod, Ken. 1982. A defense of the limit assumption. *Philosophical Studies* 42. 53–66.
- Winter, Yoad. 2025. Karttunen logic for presupposition projection. *Linguistics and Philosophy* 48. 411–418.
- Yalcin, Seth. 2007. Epistemic modals. *Mind* 116. 983–1026.
- Zimmermann, Thomas Ede. 2000. Free choice disjunction and epistemic possibility. *Natural Language Semantics* 8. 255–290.

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